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LRSM $WR \rightarrow tb$ Channel Analysis

Decay channel: $WR \rightarrow e/\mu + N \rightarrow e/\mu + WR^* \rightarrow tb$

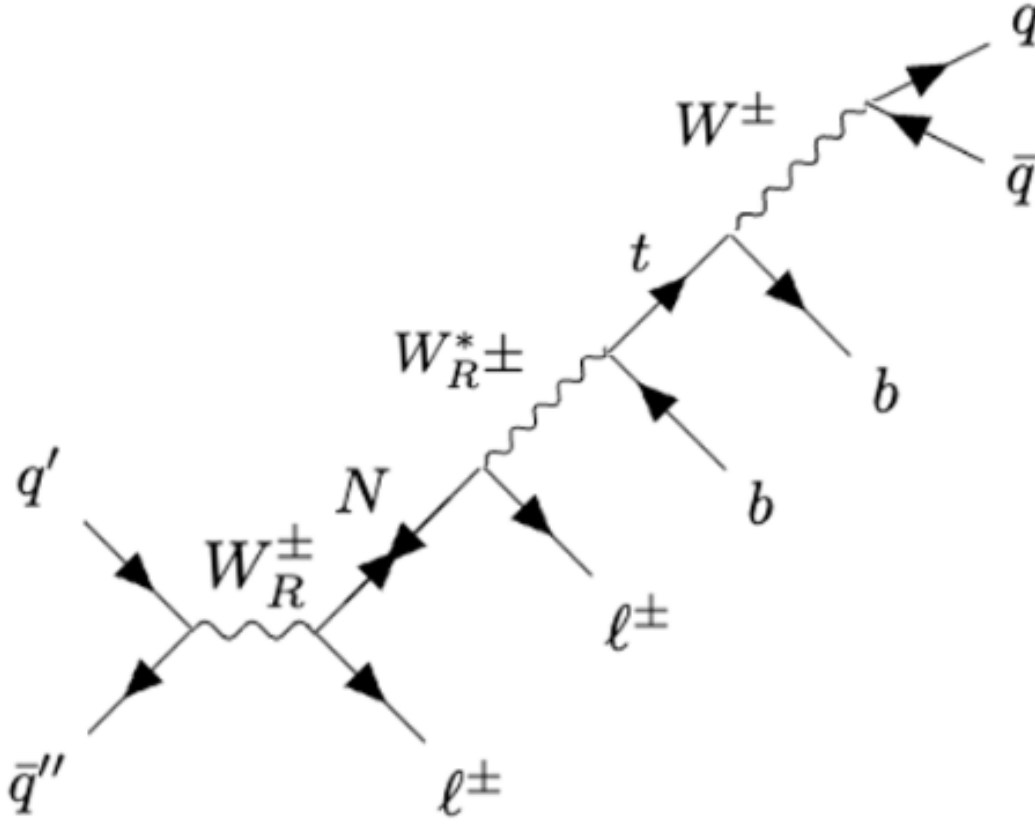


Figure 1: Figure 1: Channel diagram

1. Introduction

1.1 Motivation and Channel Selection

Previous research focused on the **WR high, N low** region where N particles are highly boosted, causing jets to cluster together. The primary decay modes studied were $WR^* \rightarrow ud, sc$ channels, which produce at most 3 substructures (q, q, τ) within a single AK8 jet.

In this channel, due to using 2 lepton and top tagging and b tagging also with high m_{ll} , we can reduce lot of backgrounds and also improvements in $WR \sim N$ region in the previous study.

Motivation For t/b Channel

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A novel search strategy for right-handed charged gauge bosons at the Large Hadron Collider

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ABSTRACT: We explore the potential of the Large Hadron Collider (LHC) in detecting a signal originating from the production of a heavy $SU(2)_R$ charged gauge boson that then decays into a top-bottom quark pair via the mediation of a right-handed neutrino, $pp \rightarrow W_R \rightarrow N_R \ell \rightarrow \ell t b$. Such a channel, that we study in the context of the minimal Left-Right Symmetric Model, contrasts with conventional smoking-gun signatures targeted experimentally and phenomenologically in which only light quarks are involved. We propose a selection strategy aimed at extracting such a top-bottom signal and we estimate the resulting sensitivity of the LHC to the model. Our results demonstrate the potential impact of such a search and we therefore urge the experimental collaborations to carry out a similar analysis in the light of present and future data.

KEYWORDS: Left-Right Models, Jets and Jet Substructure, Higher-Order Perturbative Calculations

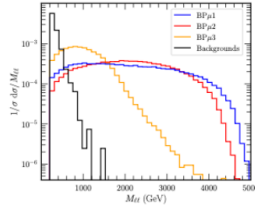
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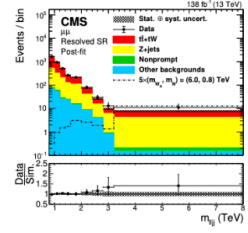
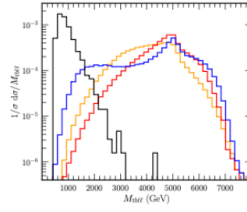
JHEP02(2024)150

The aim of this study is to propose a novel search strategy for the production of an $SU(2)_R$ charged gauge boson W_R and a neutrino N_R at the LHC, relying on the $W_R \rightarrow N_R \ell \rightarrow \ell t b$ decay chain with the W_R emerging from charged current Drell-Yan (DY) production. The top quark, being produced from a heavy neutrino decay, is highly boosted in most of the cases. **Jet substructure methods and top taggers have therefore the potential to efficiently reduce the SM background**, together with specific kinematics variables exploiting the richness of the final state. The use of this channel can thus be crucial not only for discovery purposes but also for diagnostics as a probe of the properties of the mLRSM as a whole.



BPp1 : W_R 5100, N_R 1020
BPp2 : W_R 5100, N_R 2550
BPp3 : W_R 5100, N_R 4700

Backgrounds: $t + X$, $t\bar{t} + X$



[https://doi.org/10.1007/JHEP04\(2022\)047](https://doi.org/10.1007/JHEP04(2022)047)

Figure 2: Figure 2: Previous study motivation

Previous study data (EXO-20-002)

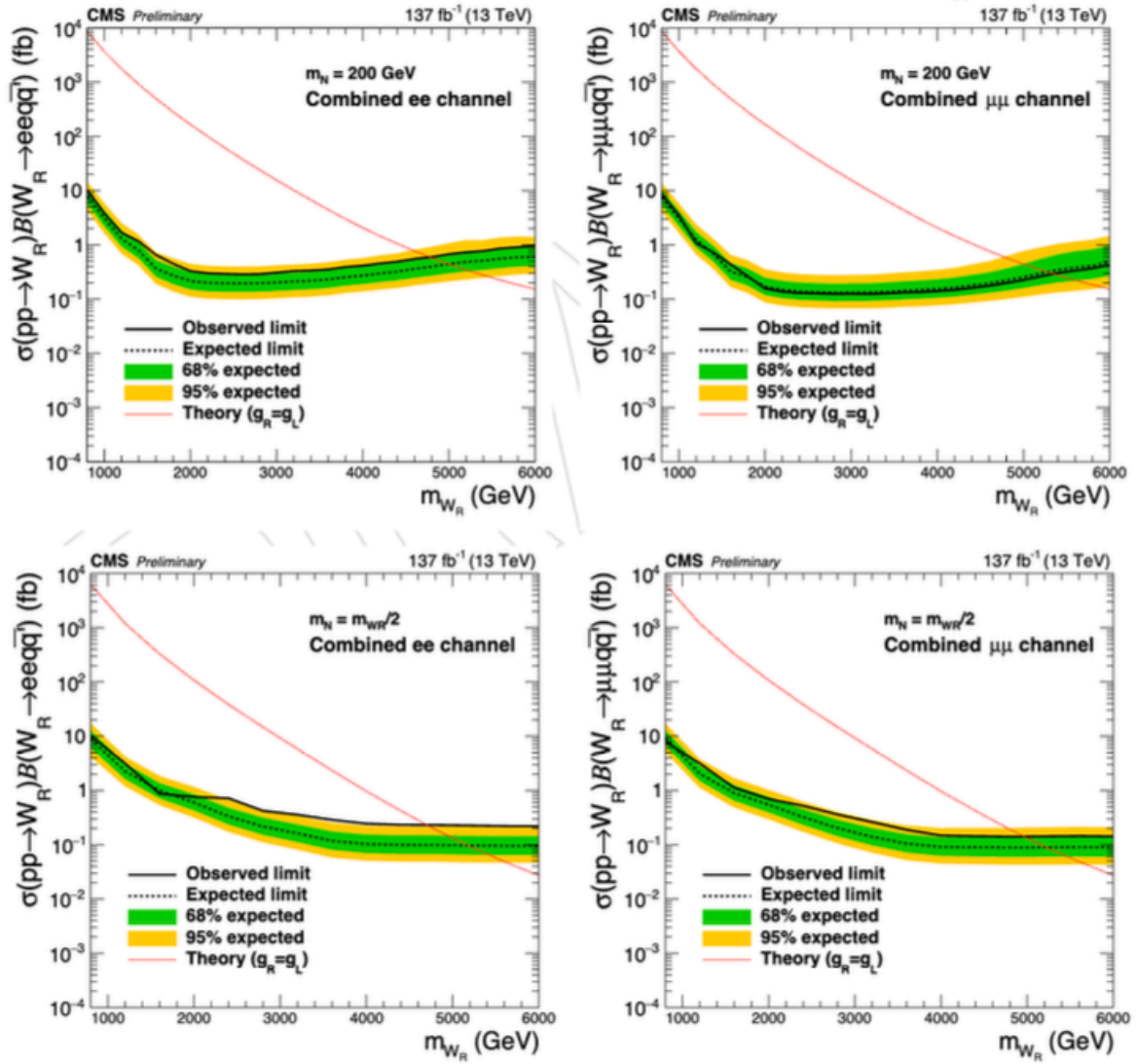


Figure 3: Figure 3: Cross-section comparison

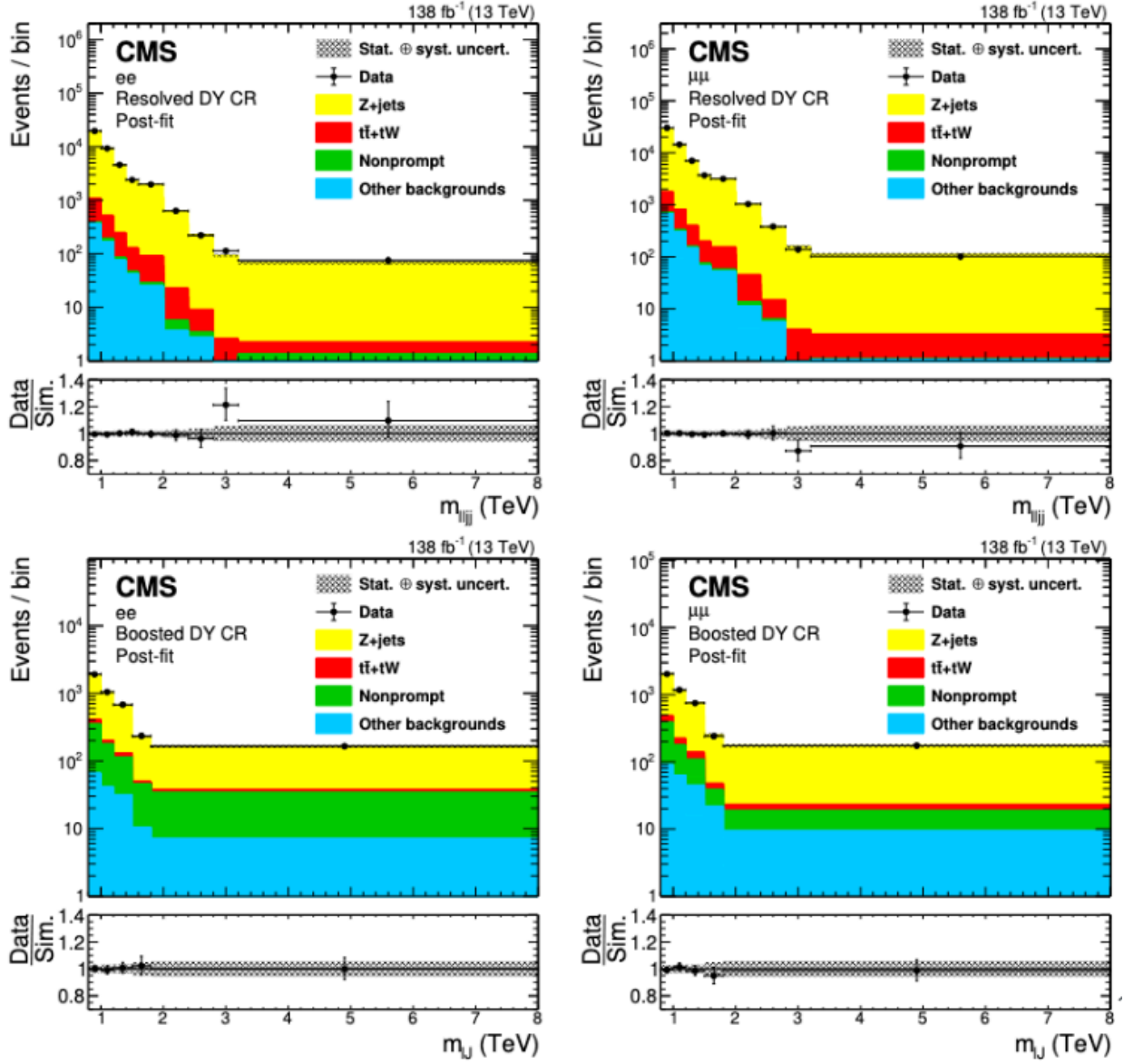


Figure 4: Figure 4: Mass limits from previous analysis

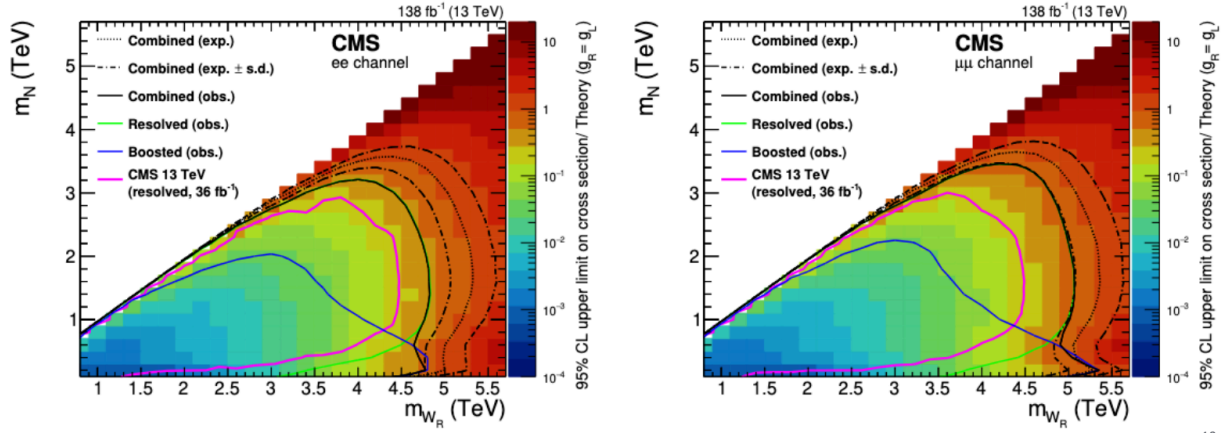


Figure 5: Figure 5: Exclusion contours

$]M_{\ell\ell jj}^{\min}, M_{\ell\ell jj}^{\max}]$	Electron channel		Muon channel	
	Data	Background	Data	Background
]800, 1000] GeV	1106.0	1103.5 ± 26.607	1639.0	1670.7 ± 39.774
]1000, 1200] GeV	646.0	631.51 ± 16.968	946.0	925.99 ± 23.917
]1200, 1400] GeV	332.0	323.23 ± 10.736	518.0	500.33 ± 14.869
]1400, 1600] GeV	170.0	169.69 ± 6.8418	268.0	263.88 ± 9.3498
]1600, 2000] GeV	143.0	157.55 ± 9.505	216.0	215.18 ± 8.2146
]2000, 2400] GeV	62.0	52.327 ± 3.9676	80.0	73.482 ± 4.4654
]2400, 2800] GeV	25.0	19.567 ± 1.5493	30.0	25.943 ± 2.3125
]2800, 3200] GeV	10.0	8.9907 ± 1.209	13.0	9.7557 ± 1.1603
]3200, 8000] GeV	13.0	6.2463 ± 0.77892	11.0	7.8119 ± 0.84286

arXiv:2312.08521v2

EXO 20 -002 data / bkg rates (light quark channel)

Figure 6: Figure 6: Additional results

1.2 Challenge with **tb** Channel

Extending the analysis to include the **tb decay channel** presents kinematic challenges:

- **Top quark rapid decay:** $t \rightarrow b + W$, where W further decays to either:
 - Hadronic: $q\bar{q}$ ($\sim 70\%$ BR)
 - Leptonic: l_{ν} ($\sim 30\%$ BR)

- **Boosted regime complexity:** In the $WR \gg N$ mass regime, all decay products (3 jets from top, b quark, lepton) become highly boosted, creating 5 substructures within a single cone - making reconstruction difficult.

1.3 Proposed Solution: $WR \sim N$ Mass Region

Strategy: Target the $WR \sim N$ mass region where:

1. **N becomes resolved:** Lepton and off-shell WR^* separate
2. **Similar kinematics:** Off-shell WR^* has similar mass to N, leading to comparable momentum
3. **Separable final states:** When $WR^* \rightarrow tb$, the top and bottom quarks can be distinguished using AK4 jets
4. **T,B-tagging capability:** b quarks (from and WR^* decay) can be identified using b-tagging, and top quark can be Top tagged

1.4 Kinematic Regimes

High p_T scenario (high mass WR , low mass N): - 3 objects contained within single cone - Boosted top jet reconstruction

Low p_T scenario ($WR \sim N$ mass region): - WR^* cone splits into top and b cone - Resolved reconstruction possible

Reference masses: - Top quark: ~ 173 GeV - Bottom quark: ~ 4 GeV - W boson: ~ 80 GeV (provides moderate boost to decay products)

1.5 Analysis Variables and Cuts

Key invariant masses: - $M_{\{ll \rightarrow tb\}}$: Full system invariant mass - $M_{\{ll\}}$: Dilepton invariant mass (peaks at ~ 90 GeV for Z resonance)

Selection considerations: - Fat jet p_T cut at 200 GeV may handle lepton selection - In boosted regimes, leptons correlate with jet boost $\rightarrow$ additional lepton cuts may be needed - M_{ll} particularly useful for boosted leptons due to Z peak structure - More study needed

1.6 Top Decay Complications in LRSM

Chirality constraints: Since q_T originates from W_R^* with right-handed chirality, all subsequent decay products maintain right-handed chirality. This leads to $t \rightarrow W_R b_R$ instead of the SM decay $t \rightarrow W_L b$.

Key challenges:

1. **Non-SM decay mode:** Need to account for $W_R b_R$ decay channel separately
2. **Off-shell W_R suppression:** Since W_R decays off-shell near top mass, cross-section significantly reduced for massive W_R . Leptonic channels become negligible.
3. **Chirality flip probability:** For W_L production, chirality flip required with probability $\sim m_T/E_T$
 - Example: $3 \text{ TeV } W_R \rightarrow t(1.5 \text{ TeV}) + b(1.5 \text{ TeV}) \rightarrow <10\% \text{ flip probability}$
 - Results in SM-like W boson production

Mass regime dependencies:

- **Low W_R (top-resolved):** Both W_L and W_R decay modes possible with finite probabilities
- **High W_R (top-boosted):** Predominantly W_R decay, difficult to observe due to off-shell nature

Alternative approach: In high W_R , low N region, if initial W_R is off-shell, reduced top boost enables W_L decay mode.

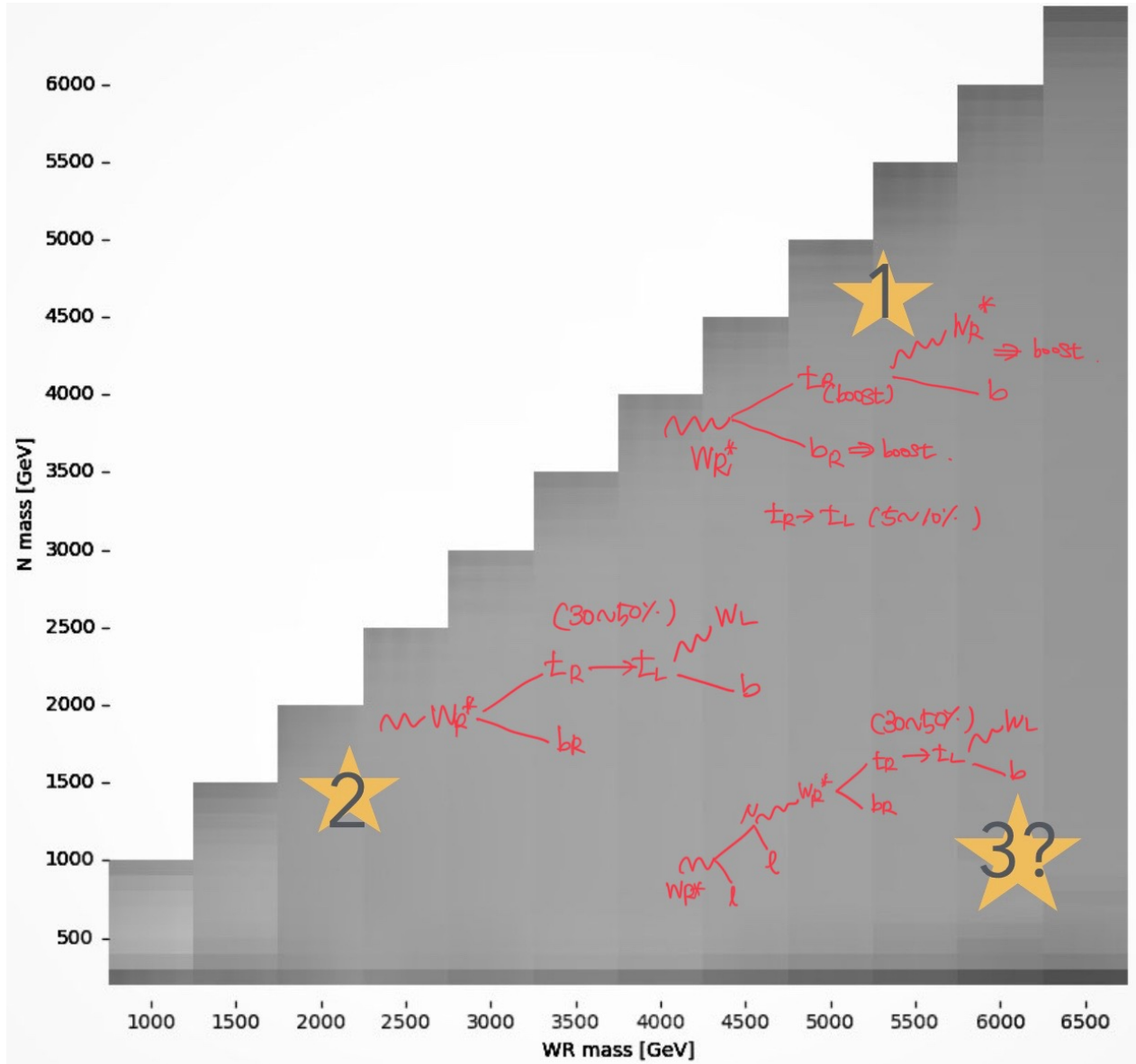


Figure 7: Figure 7: Kinematic diagram

MadGraph simulation setup: - Top couples to both WR and WL: $t \rightarrow (WR + WL)b$ vs $t \rightarrow WL b$ - Off-shell $WR \rightarrow qq$: $t \rightarrow (qq)b$ process - Comparative study: $(WL + WR)$ vs (WL) only - Investigation of automatic chirality flip handling in LRSM

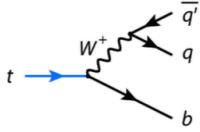
Results: - Combined $(WL + WR)$ shows $\sim 1\%$ higher cross-section than WL only

Conclusion: WL production dominates across most parameter space (off-shell WR

production less favorable than chirality flip) - Including tb channel increases cross-section by $\sim 10\%$ compared to standard approach

Decay width comparison (3 TeV WR): - generate $t \rightarrow q q b / w+$: Width = $1.1 \times 10^{-8} \pm 1.4 \times 10^{-11}$ GeV - generate $t \rightarrow q q b / wr+$: Width = 0.97 ± 0.0007 GeV - **Result**: WR decay channel negligible at high masses

Right handed top?



How can right handed top can decay to SM W?

1. $t_R \rightarrow t_L$
2. Just decays to right handed W

Due to lot of mass difference with t_R & W_R^* $\rightarrow$ highly suppressed $\rightarrow$ "will be almost decay to $t_L \rightarrow W_L$ "

Checked with cross section in 2 Cases [LO]

```
import model EffLRSM_NLO
define p = g u c d s u~ c~ d~ s~
define quark = u c d s u~ c~ d~ s~
define j = p
define ll = e+ e- mu+ mu-
define nn = n1 n2
define bot = b b~
define top = t t~
define wr = wr wr~
generate p p > nn ll, (nn > ll top bot, (top > quark quark bot))
```

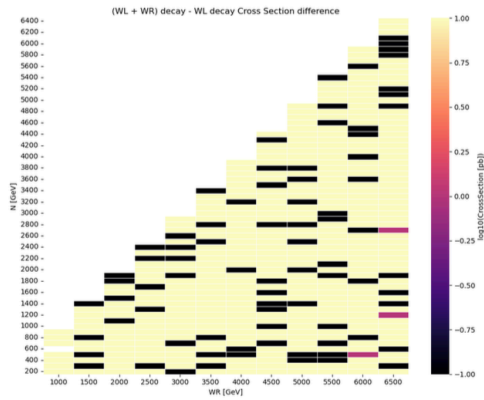
```
import model EffLRSM_NLO
define p = g u c d s u~ c~ d~ s~
define quark = u c d s u~ c~ d~ s~
define j = p
define ll = e+ e- mu+ mu-
define nn = n1 n2
define bot = b b~
define top = t t~
define wr = wr wr~
define w = w w~
generate p p > nn ll, (nn > ll top bot, (top > w bot, (w > quark quark )))
```

On 2025-06-17 04:57, An Chihwan wrote:
 I think it's right.
 * I hope this message finds you well.
 * My name is Chihwan An, and I am a first-year master's student in the CMS group at Seoul National University. I was told by Shihyun Jeon who was in contact with you for other heavy resonance publications that the SHG group worked on during last few years to seek your insights on some questions I have. I am currently studying Left-Right Symmetric Models (LRS), and have been making use of the LRS-CP model that you and your collaborators developed. I encountered one question during a study and unfortunately those problems were not explained explicitly in following website and paper:
 * <https://arxiv.org/abs/1508.05011> [1].
 * <https://arxiv.org/abs/1710.08085> [2].
 * I am currently studying the process $qqqq \rightarrow W_R^+ W_R^- \rightarrow W^+ W^- + t \bar{t}$, followed by the top quark decay $t \rightarrow W b$ where the top quark is expected to be produced via a right-handed WR interaction. I would greatly appreciate your clarification on two points regarding the chirality and decay treatment of the top quark in the UFO implementation:
 * Our question is the following. With the current model file taken into account that $t_R \rightarrow W_R^+ b$, R implies that W_R has to be off-shell since our interest is when $W_R \rightarrow W_L$ and generate $t \rightarrow W_L b$ decay branching ratio? I understand that the mass suppression will be much larger for W_R but on the other hand, due to the preference of right-handedness of top (implying the left-hand top suppression $\sim m/E \ll 1$), we weren't sure (a) if this is really ignorable and (b) if the model file accounts for these effect already.
 * If $MWR > m_t$, then the 2-body decay process $t \rightarrow W_R b$ is "most" possible as the kinematically forbidden.
 * If instead you simulate the full 3-body decay processes $t \rightarrow b W^+ W^-$ where W^+ and W^- are any SU(2) pair of fermions (le, ν , ν , ν , ν , etc), then both WSM and WR will be included.
 * If you simulate, for example, the channel $t \rightarrow W^+ W^- b$, then you will have two diagrams: one with $t \rightarrow W^+ W^- b$ and one with $t \rightarrow W^+ W^- b$. Full spin correlation and off-shell effects are taken into account. Just be aware that the $t \rightarrow b W^+ W^-$ process will rely on the widths of WSM and WR.

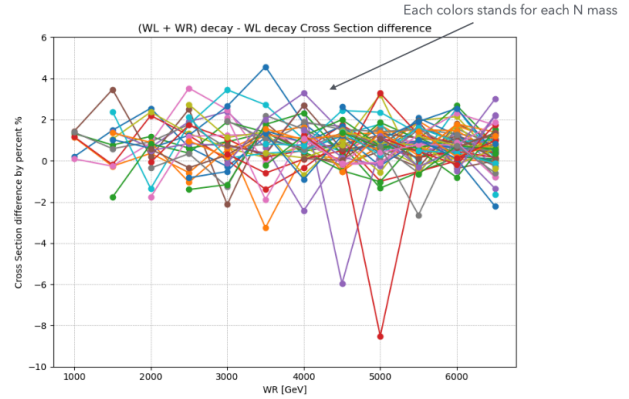
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Figure 8: Figure 8: Cross-section comparison WL vs WR

Right handed top?



Yellow region is when $W_L + W_R$ decay has more cross section
Black region is when W_L decay has more cross section



Average $\sim 0.7\%$ (which means W_R contributes cross-section about 0.7%)
-> It would not be a big problem to consider W_R topology..

23

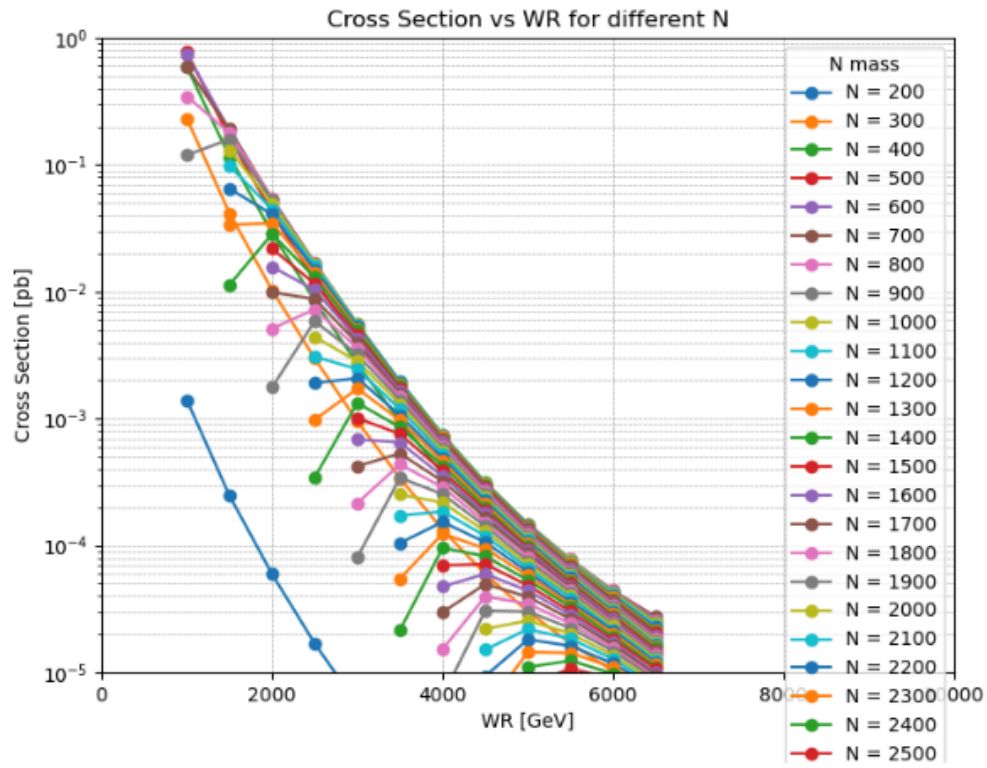
Figure 9: Figure 9: Cross-section analysis

Checked with WL + WR both decay mode and the case with decays with only SM W. There was no significance pattern when does WR contributes to cross section.

Yellow region means when WL + WR decays it has more cross section and it was order of less than 5% in LO. Also black region is when WL + WR makes less cross section and DONT KNOW WHY yet why this happens, also in LO.

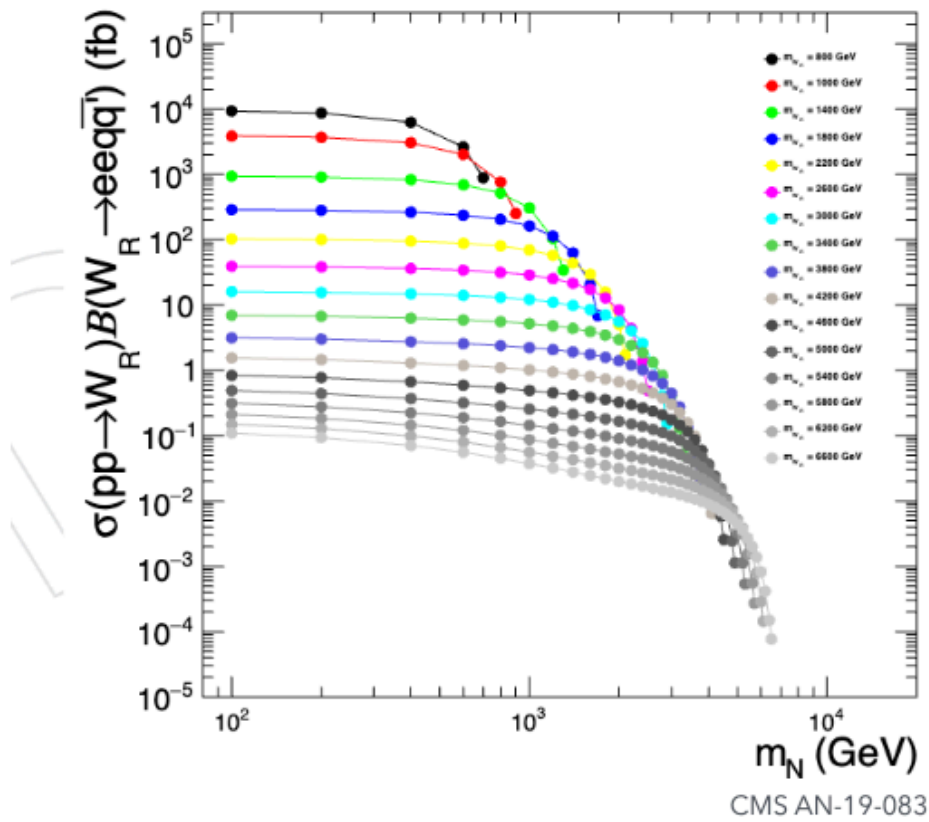
1.7 Cross-Section Studies

Cross section is checked with LO gridpacks.



- Cross section up to 6.5 tev (t/b jet channel)

Figure 10: Figure 10: Cross-section gridpack results



- Cross section up to 6.6 tev (light jet channel)

Figure 11: Figure 11: Cross sections for exo 20-002

Cross sections for exo 20-002

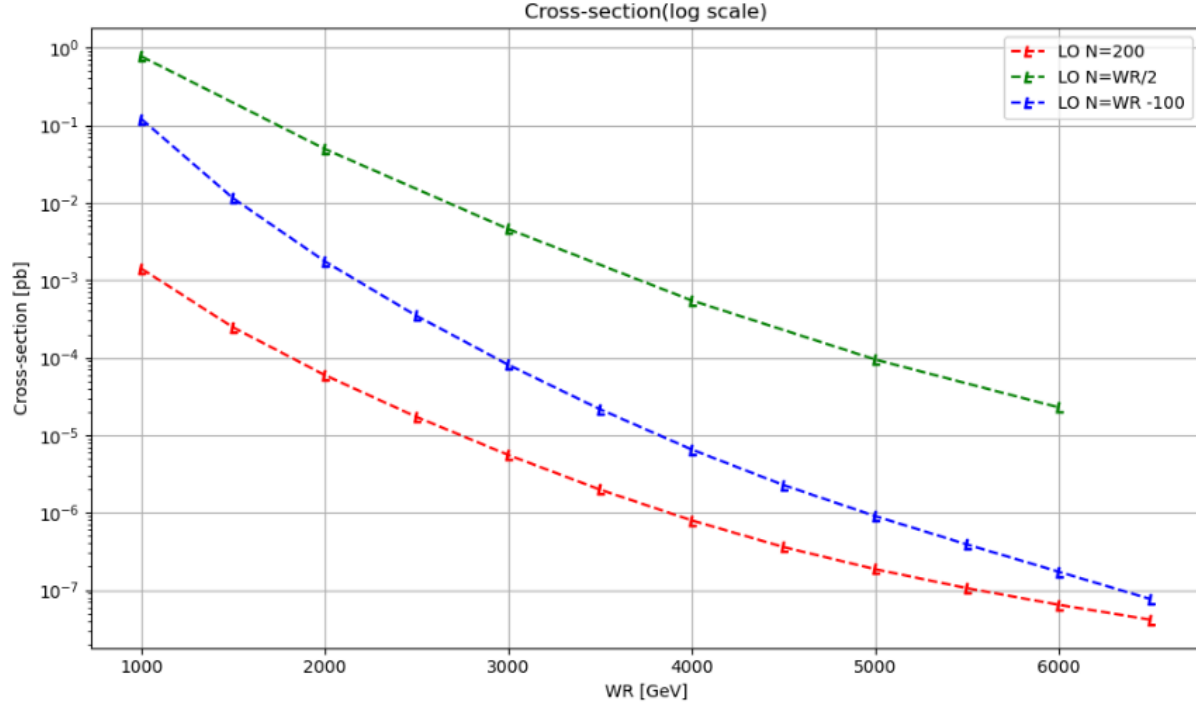


Figure 12: Figure 12: Cross-section detailed analysis

Cross section when N is $\sim WR/2$ has the most cross section, because $N = 200$ it is constraint due to top mass, and also $N \sim WR$ has no much kinematic phase space to decay.

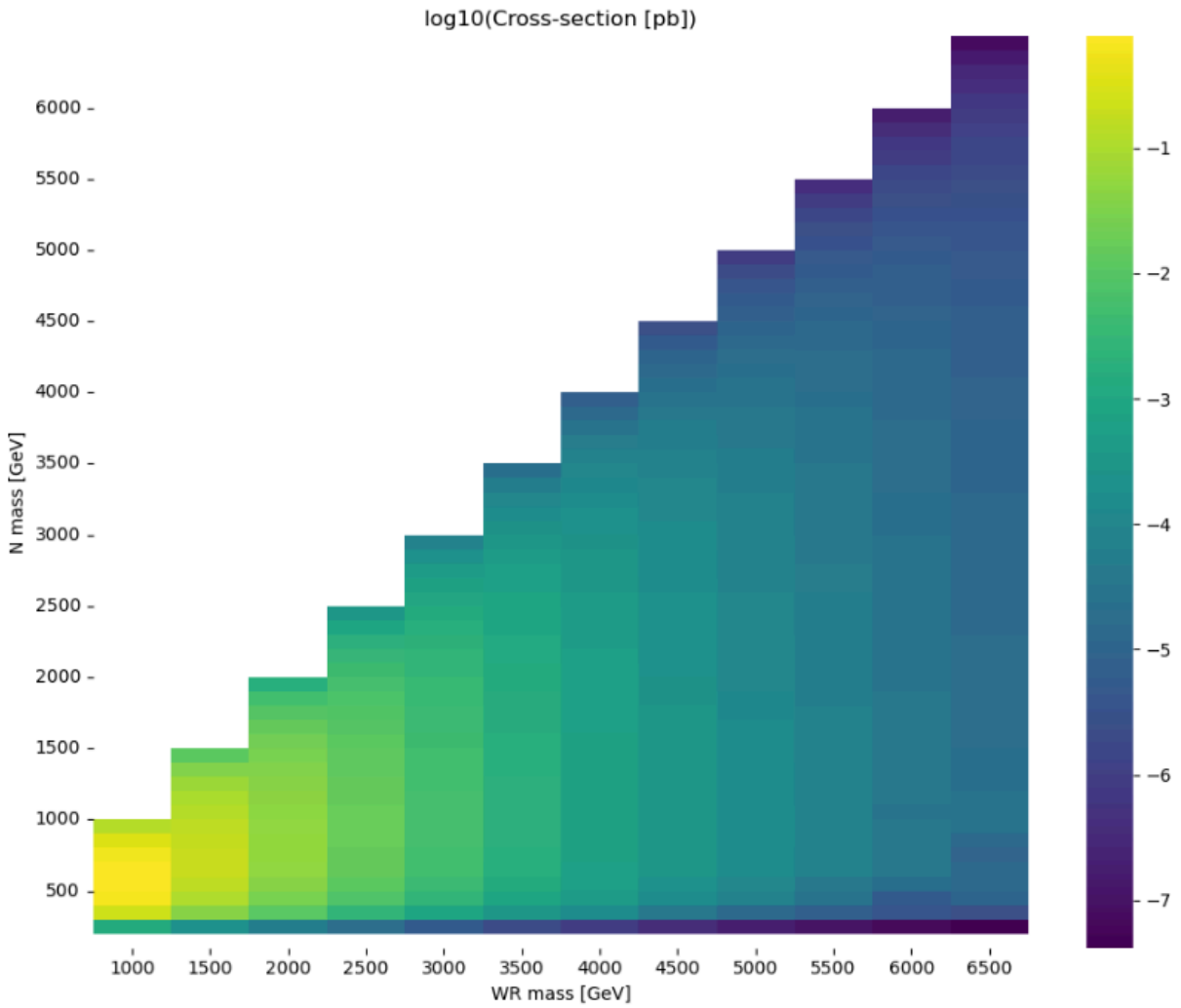


Figure 13: Figure 13: Full cross section

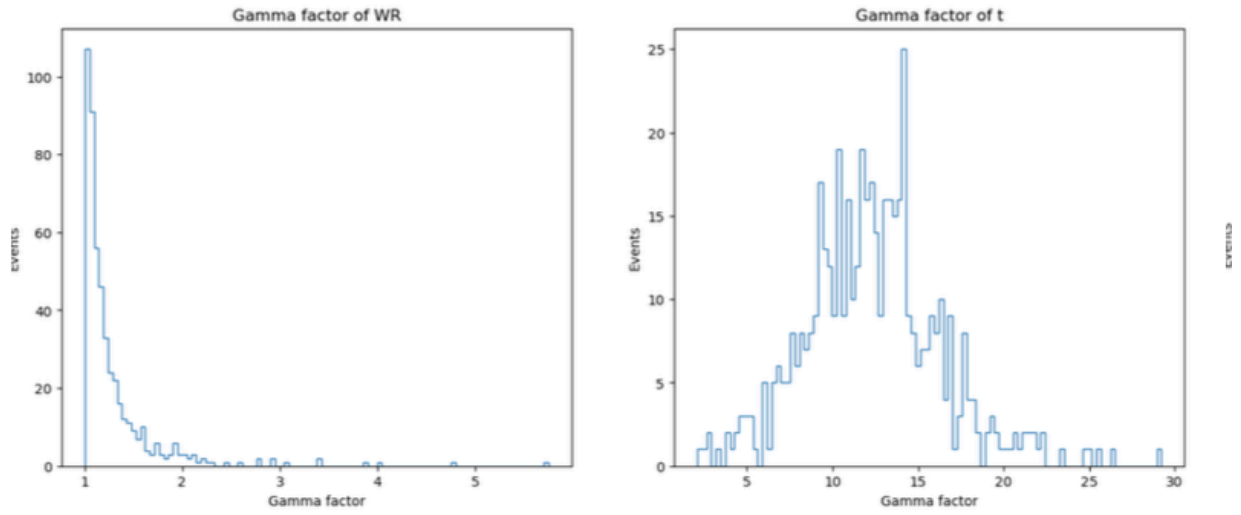
Full cross section

2. Signal Analysis

2.1 Kinematics and Event Topology

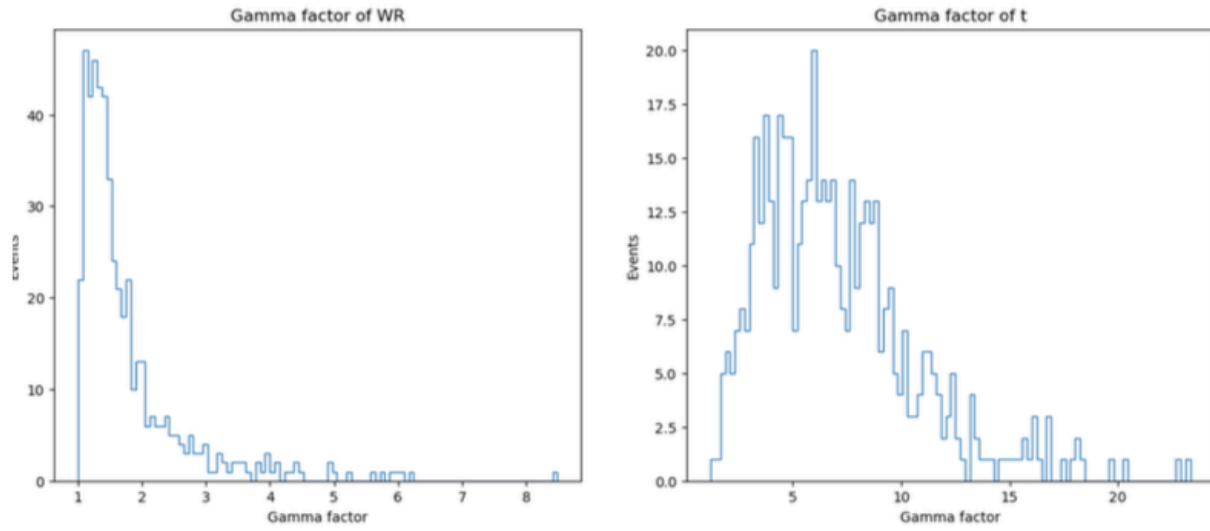
Analysis approach: Event kinematics studied using LHE files (parton level) to understand ΔR and p_T distributions.

Boost Factors for non observable objects (WR, top)



W_R 5000 , N 4900

Figure 14: Figure 14: Boost factors WR

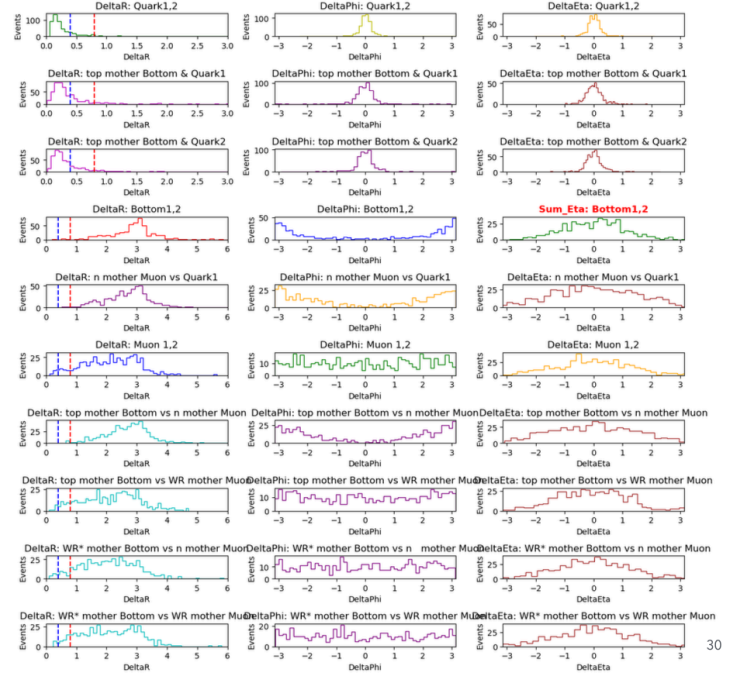


$$W_R 5000 , N 2500$$

Figure 15: Figure 15: Boost factors top

Delta R, Delta eta, Delta phi for each variables

ΔR , $\Delta\eta$, $\Delta\phi$
For $W_R 5000 N4900$



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Figure 16: Figure 16: Angular correlations

ΔR , $\Delta\eta$, $\Delta\phi$

For $W_R 5000$ $N2500$

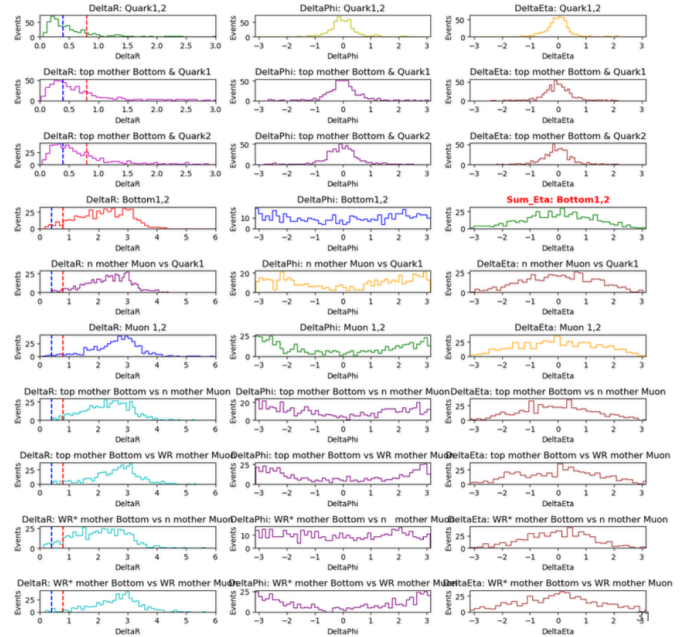


Figure 17: Figure 17: Additional angular distributions

2.1.1 WR ~ N Mass Region (5000, 4900 GeV) Key observations:

- **WR* rest frame:** t and b quarks produced back-to-back (ΔR between bottom quarks ~ 3)
- **Limited boost:** Since WR* not highly boosted, tb quarks separate significantly
- **Lepton correlations:**
 - l1 (from WR): Can appear in all directions
 - l2 (from N): Back-to-back with WR* in N rest frame
- **Unexpected behavior:** Top quark less boosted than expected, follows WR* direction
- **Bottom quark:** Highly boosted, random direction relative to l2
- **Interpretation:** Suggests moderate WR* boost regime

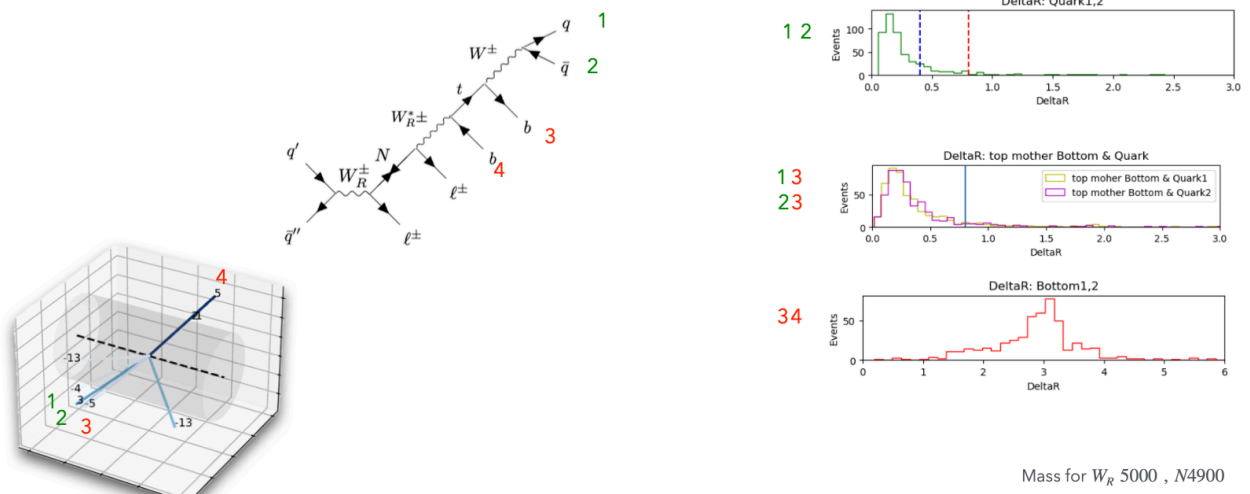


Figure 18: Figure 18: Kinematic analysis 4900 GeV case

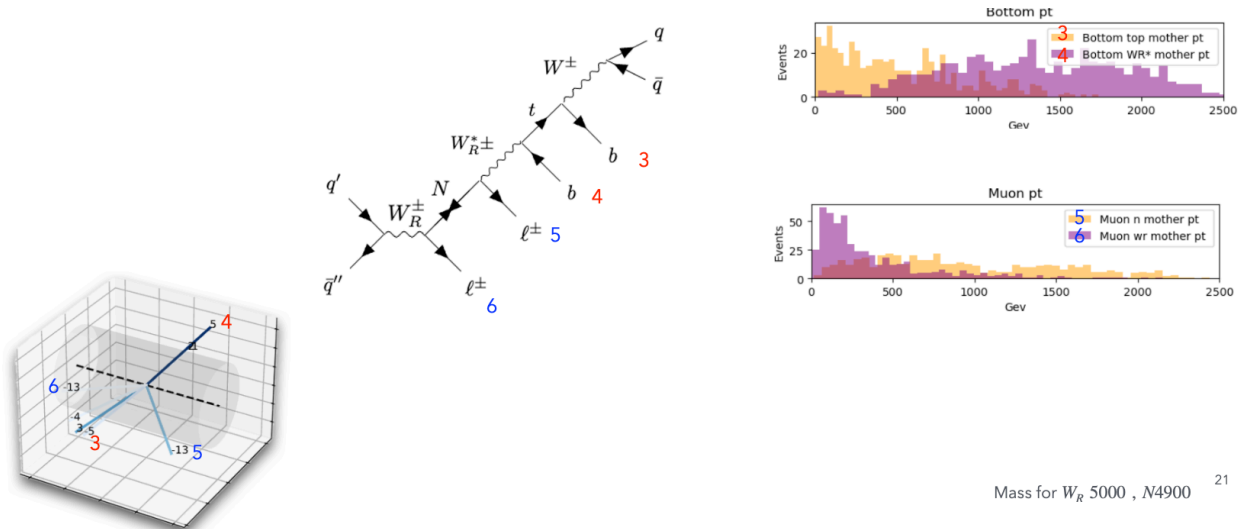


Figure 19: Figure 19: Additional kinematic plots

2.1.2 $W_R \sim N/2$ Mass Region (5000, 2500 GeV) Distinct topology observed:

- Initial state:** N and $l1$ produced back-to-back, with N significantly boosted
- N decay dynamics:**
 - $l2$ partially opposes N boost $\rightarrow$ moderate displacement from W_R^* direction
 - Some $l2$ events still follow W_R^* direction due to boost competition
- W_R^* behavior:** Inherits N direction due to boost inheritance
- Final state tb decay:**
 - b

quark: Receives strong additional boost $\rightarrow$ significant separation from top jet - **Top jet:** Moderately boosted (reduced WR^* mass) $\rightarrow$ partially follows N direction - **Angular correlation:** b quark direction depends on decay orientation: - Opposite WR^* direction $\rightarrow$ back-to-back with system - Same WR^* direction $\rightarrow$ collinear with top jet

2.2 AK8 Top Jet Analysis

Primary objective: Identify and characterize signal AK8 jets while understanding potential backgrounds and misidentification sources.

Analysis strategy: 1. **Signal jet identification:** Match closest AK8 jet to generator-level top quark 2. **Background characterization:** Study distance distributions and top scores for nearby AK8 jets 3. **Selection validation:** Test discriminating power of top tagger and soft-drop mass cuts 4. **Efficiency assessment:** Evaluate performance with randomized AK8 jet selection

2.2.1 Signal Efficiency and Fake Jet Studies **Objective:** Assess signal efficiency by identifying potential fake signals that mimic top jets.

Methodology: 1. **Geometric matching:** Examine ΔR between post-FSR generator-level top quarks and reconstructed AK8 jets 2. **Discriminant analysis:** Evaluate soft-drop mass and top tagger scores for nearby AK8 jets to assess misidentification probability

Results for $WR = 5000$, $MN = 4900$ GeV:

Key findings: - **Clean separation:** No overlap with other fat jets in signal region - **Low-mass artifacts:** Missing entries at low soft-drop mass due to poor jet reconstruction - **B-jet leakage signature:** Soft-drop mass range 80-150 GeV corresponds to top jets missing b-quarks - **Mass dependence:** $WR \sim N/2$ shows more b-quark leakage than $WR \sim N$ due to reduced WR^* boost $\rightarrow$ less collimated decay products

2.2.2 Signal Selection Efficiency **Overall performance assessment:** Application of top tagger and soft-drop mass cuts to complete signal sample

Efficiency results: 499/500 events (99.8%) successfully identify signal top quarks with AK8 jets

Mass point comparisons:

$WR = 5000$, $MN = 4900$ GeV:

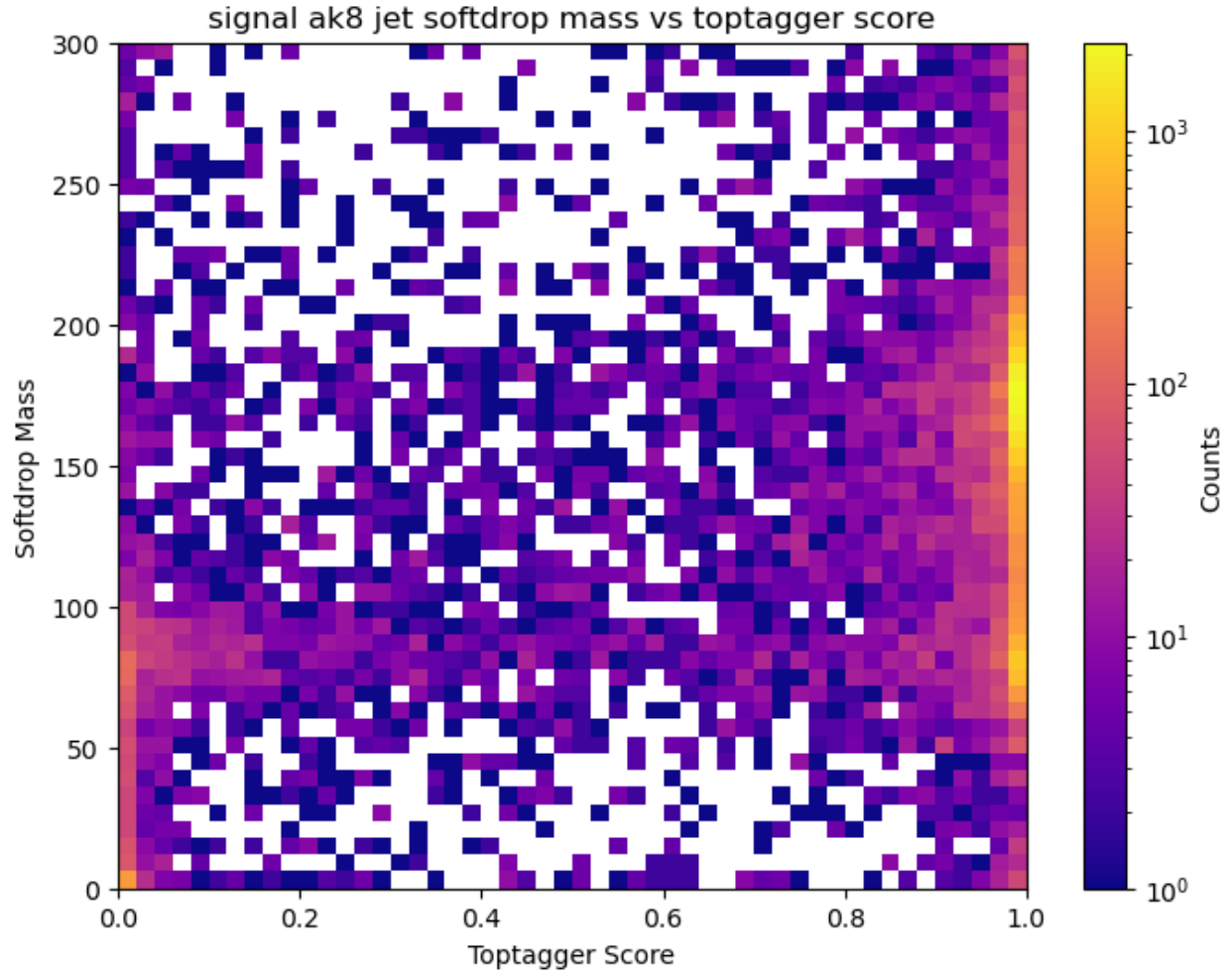


Figure 20: Figure 20: Top tagging efficiency - high N mass

2.2.3 Fake Jet Background Assessment Random AK8 selection test: Evaluate false positive rate when applying top tagging and soft-drop mass cuts to arbitrary AK8 jets

Key findings: - **Primary signal identification:** Closest ΔR match reliably identifies true signal - **Secondary contamination:** Some jets within $\Delta R < 0.8$ pass selection cuts

2.2.4 Fake AK8 Jet Classification Event categories: No events with 3+ top-tagged jets observed

Classification scheme: 1. **Single AK8:** One jet passes selection 2. **Double AK8:** Two jets pass selection

Analysis for $WR = 5000$, $MN = 4900$ GeV **Double AK8 scenarios:** - **2S (two signal):** Not possible $\rightarrow$ no additional AK8 jets within $\Delta R < 0.8$ of signal - **Event classification:** 1. **1F1S:** One jet within $\Delta R < 0.8$ + one jet outside $\Delta R > 0.8$ pass cuts 2. **2F:** Closest jet fails cuts + two jets outside $\Delta R > 0.8$ pass cuts

Event Type	2F (two fake)	1F1S (one fake + signal)
2 AK8 jets	Low rate	Moderate rate

Single AK8 scenarios:

Event Type	1F (one fake)	1S (signal only)
1 AK8 jet	Low rate	Dominant

Key observations: - **Single AK8 events:** No issues, dominant contribution with clear signal identification - **Double AK8 events:** Lower rate but challenging discrimination $\rightarrow$ top tagger shows limited separation, soft-drop mass distributions similar to genuine top jets

Angular Analysis and Interpretation **Hypothesis:** Double AK8 events may result from b-quark/top-quark angular correlation creating secondary jets outside $\Delta R > 0.8$

Investigation approach: 1. **LHE-level analysis:** Study η , ϕ separations between b quarks from different sources 2. **Reconstructed jet correlations:** Examine η , ϕ differences between AK8 jets and generator-level tops 3. **Mass correlation:** High soft-drop masses potentially due to $WR^* \rightarrow b$ contamination 4. **Back-to-back topology:** Study large ΔR events with opposite ϕ values

Key questions: - How often do $WR^* b$ quarks fall outside $\Delta R = 0.8$ cone? - What fraction of single AK8 events contain hidden b quarks? - Are double AK8 events predominantly from b-quark leakage scenarios?

Conclusions from angular analysis: - **η difference/sum limitations:** Near-zero values make it difficult to distinguish back-to-back vs collinear configurations - **Dominant pattern:** Sum distribution peaks near 0, with very few events beyond 3 $\rightarrow$ confirms predominantly back-to-back topology - **Rare collinear cases:** Small fraction

shows same-direction propagation - **Fake jet mechanism:** Collinear events likely create additional $\Delta R \sim 0.8$ cones from W-boson (from top) + b-quark (from WR^*) combinations - **Event rate consistency:** No more than 2 AK8 jets pass baseline cuts, consistent with this interpretation

2.2.5 AK8 Jet Cleaning and Efficiency Losses For cleaning strategy, it must not overlap with signal muons that is selected and should not be overlapped with other ak8 jets. Did not consider overlapping with ak4 because it also can be reco with AK4.

After cleaning, top pt:

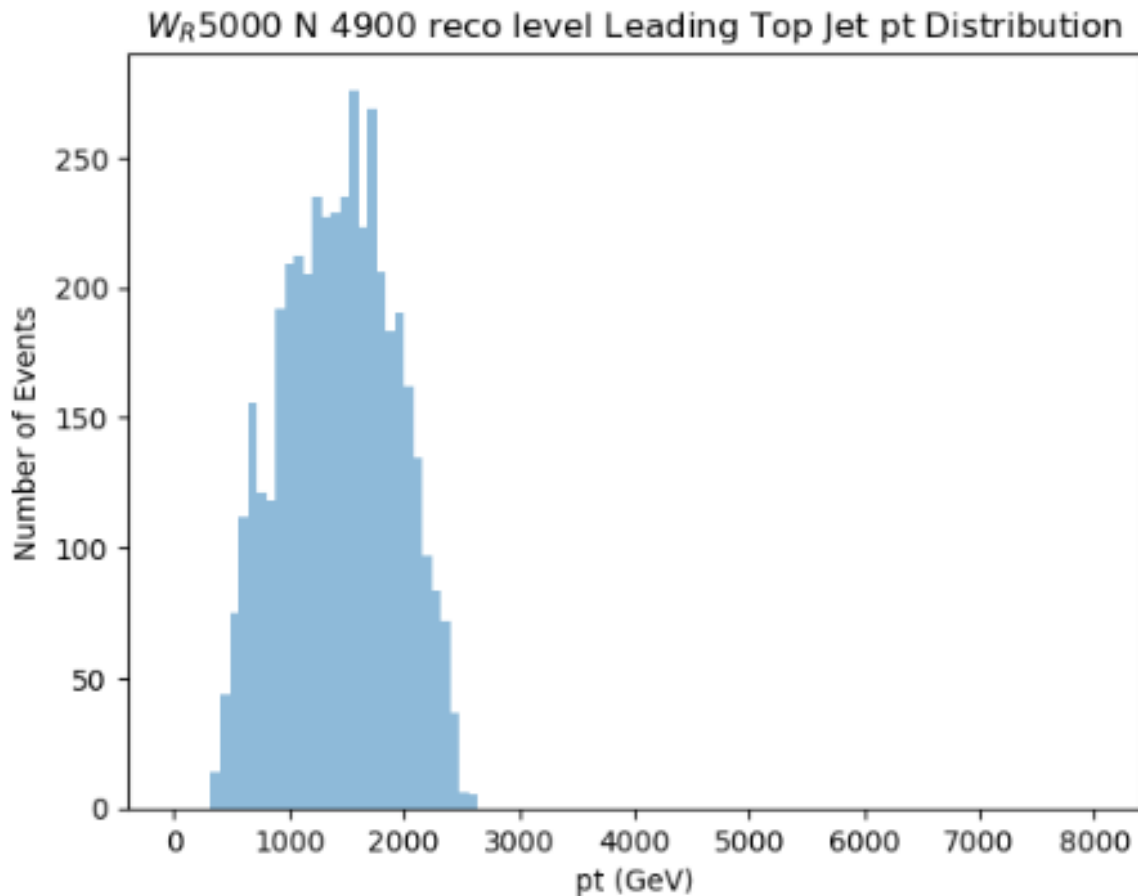


Figure 21: Figure 21: Top pT after cleaning

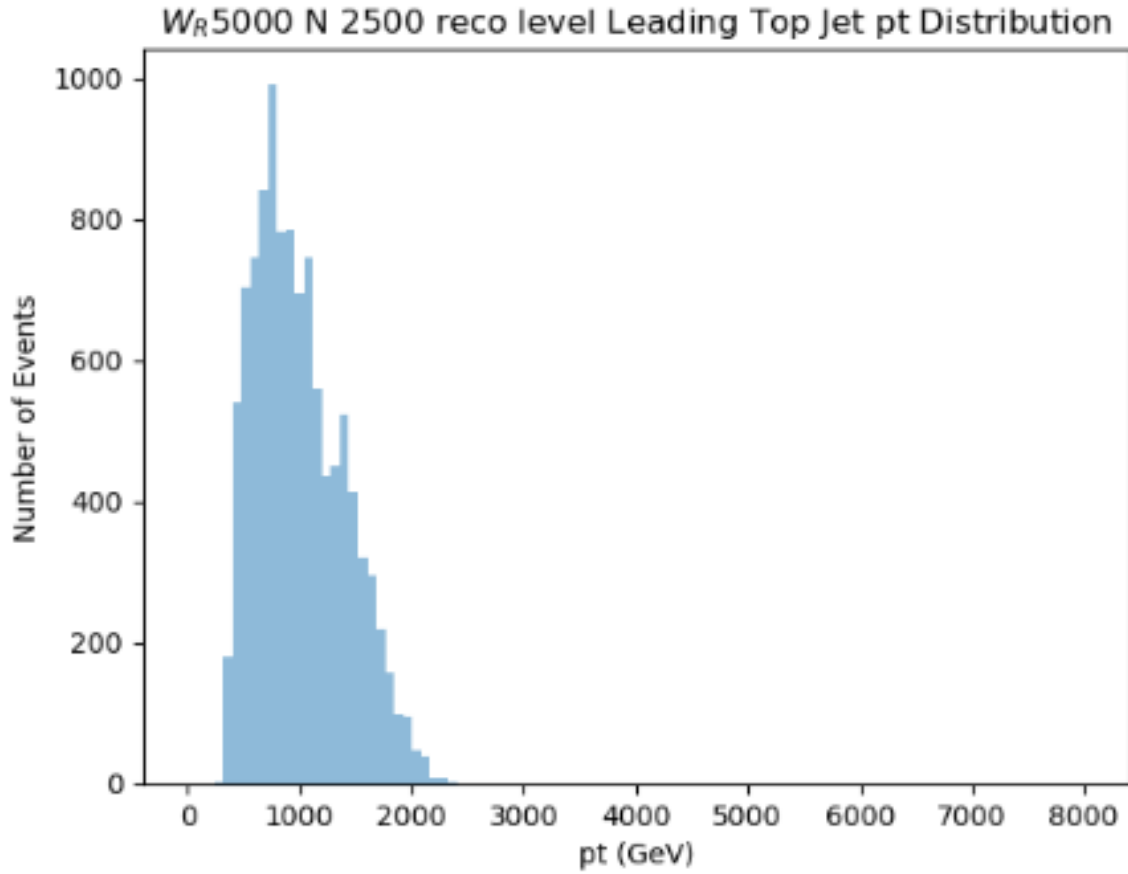


Figure 22: Figure 22: Additional cleaning results

2.2.5.1 B-Jet Cross-Contamination Analysis **Issue:** Back-to-back b-jets occasionally enter AK8 cones, though rate is relatively low

Method: Use LHE-level b-quarks to measure ΔR between signal b-jets and top jets for contamination assessment

Contamination rates:

WR = 5000, MN = 4900 GeV: - **WR* b-jets entering AK8:** 678/39,968 jets (1.2%)
- **Top b-jets escaping AK8:** 3,303/39,968 jets (8.26%)

WR = 5000, MN = 2500 GeV: - **WR* b-jets entering AK8:** 4,710/56,759 jets (8.3%)
- **Top b-jets escaping AK8:** 12,591/56,759 jets (22.18%)

2.2.6 Top tagging recommendation after cleaning **Reference:** CMS Top Tagging Working Points

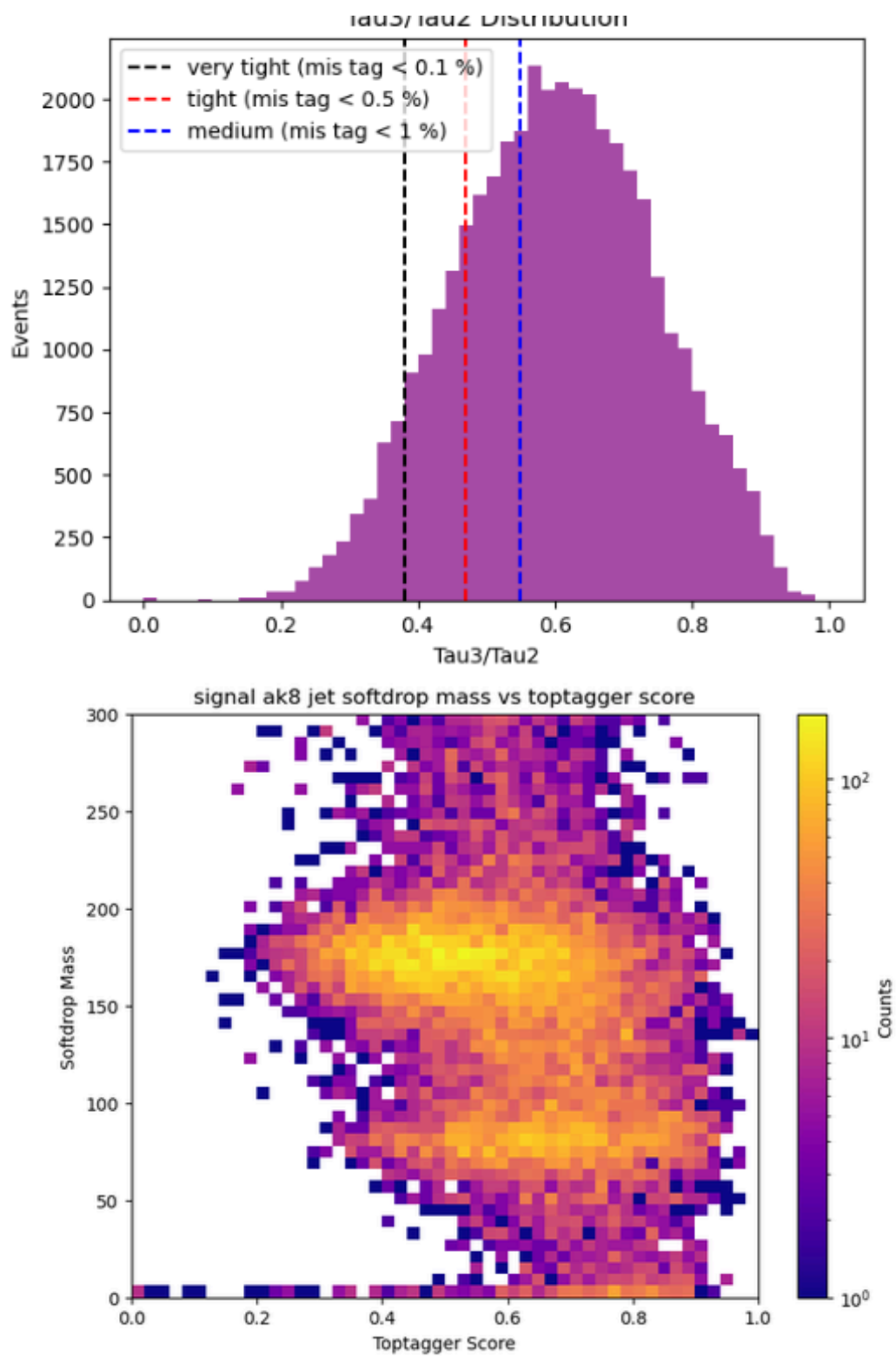


Figure 23: Figure 23₂₈ CMS Working Points

There is a WP with variable tau3/tau2 but it has low eff than particle net so It should be waiten until WP came with version NanoAOD v 15

For working process:

TvsQCD Loose WP

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- ▶ Results for 2022EE, $WP = 0.698$, $P_T \in [600, 1200] \text{ GeV}$
- $SF_{Top-Merged} = 0.92 - 0.07 / + 0.08$

Figure 24: Figure 24: Working process 1

TvsQCD Tight WP

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- ▶ Results for 2022, $WP = 0.979$, $P_T \in [300, 400] \text{ GeV}$
- ▶ $SF_{Top-Merged} = 1.10 - 0.13 / + 0.21$
- ▶ Prefit (dashed) Vs Postfit (solid) distributions

Figure 25: Figure 25: Working process 2

Top SFs

21/24

WP (Mist.Rate)	$300 \leq P_T \leq 400 \text{ GeV}$	$400 \leq P_T \leq 480 \text{ GeV}$	$480 \leq P_T \leq 600 \text{ GeV}$	$600 \leq P_T \leq 1200 \text{ GeV}$
0.98 (0.1%)	$1.10^{+0.21}_{-0.13}$	$0.83^{+0.08}_{-0.06}$	$0.86^{+0.07}_{-0.07}$	$1.04^{+0.12}_{-0.11}$
0.86 (0.5%)	$1.23^{+0.19}_{-0.15}$	$0.89^{+0.07}_{-0.06}$	$0.82^{+0.07}_{-0.07}$	$1.21^{+0.29}_{-0.18}$
0.68 (1%)	$1.22^{+0.20}_{-0.16}$	$0.89^{+0.06}_{-0.06}$	$0.84^{+0.07}_{-0.07}$	$1.27^{+0.29}_{-0.27}$

Table: 2022

WP (Mist.Rate)	$300 \leq P_T \leq 400 \text{ GeV}$	$400 \leq P_T \leq 480 \text{ GeV}$	$480 \leq P_T \leq 600 \text{ GeV}$	$600 \leq P_T \leq 1200 \text{ GeV}$
0.98 (0.1%)	$1.21^{+0.11}_{-0.10}$	$1.03^{+0.07}_{-0.07}$	$0.96^{+0.06}_{-0.06}$	$0.90^{+0.08}_{-0.07}$
0.87 (0.5%)	$1.38^{+0.15}_{-0.13}$	$1.04^{+0.51}_{-0.06}$	$1.01^{+0.06}_{-0.05}$	$0.89^{+0.08}_{-0.07}$
0.70 (1%)	$1.38^{+0.62}_{-0.12}$	$1.05^{+0.43}_{-0.05}$	$1.02^{+0.19}_{-0.05}$	$0.92^{+0.08}_{-0.07}$

Table: 2022EE

- High SFs for low P_T regions
- Stable (within error) as a function of P_T (for $P_T \geq 400 \text{ GeV}$)
- Asymmetric error for some 2022EE SFs

Figure 26: Figure 26: Working process 3

It might be better to use Loose WP due to high pt and for stable with error.

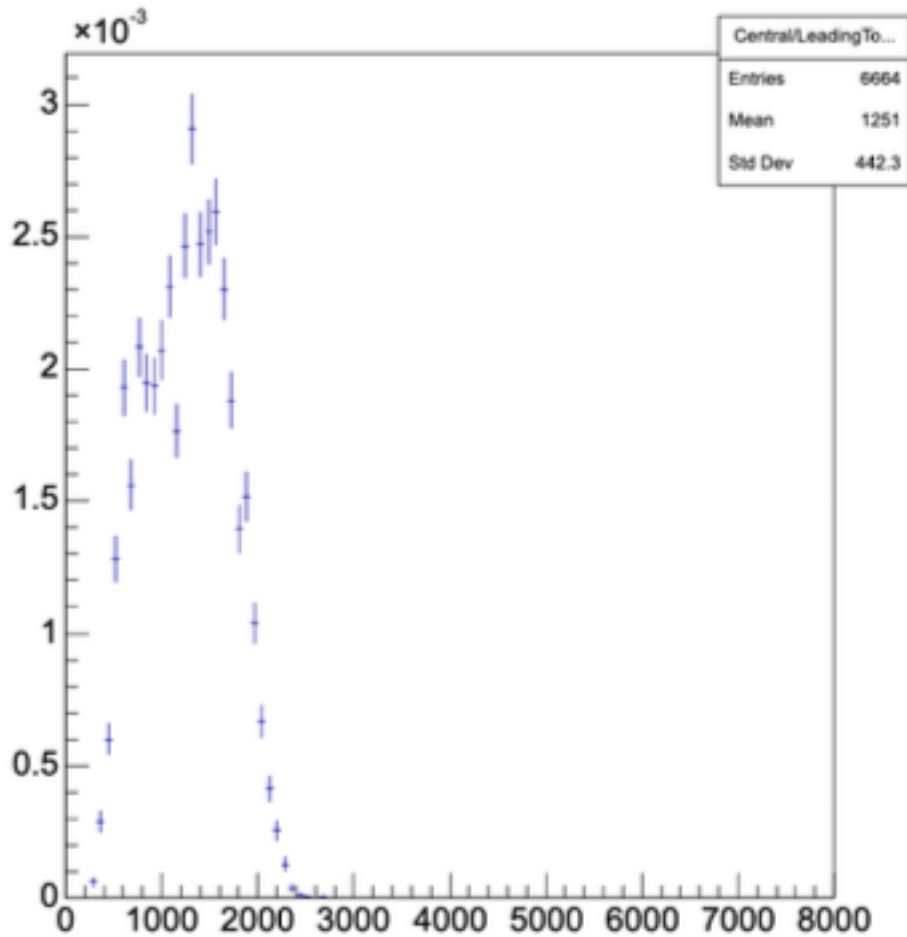


Figure 27: Figure 27: Cut adjustment

which this plot show when cut slight adjusted.

2.2.7 Conclusion When two top jets are found: If the masses are similar and they are adjacent to each other, there's no need to search for additional b-jets.

When only one top jet is present: - Case where b escapes from t and WR b doesn't enter: distinguishable by SDM - Case where b escapes from t and WR b enters: very rare, need to review if kinematically possible - Case where b doesn't escape from t and WR b enters: AK8 mass should be very large

Summary of events to discard:

AK8 Jet: 1. When top tagger 0.9, SDM 120 or higher are set as baseline (40% ~ 50%)
→ low soft drop mass cases, b escape cases 2. When two AK8s appear (when b jet protrudes forward) (5%) 3. When one AK8 appears - Among these, when $WR \cdot b$ enters (1% ~ 8%)

→ For WR 5000 N 4900 case, probability of obtaining clean and accurate top signal: ~55%
→ For WR 5000 N 2500 case, probability of obtaining clean and accurate top signal: ~40%

2.3 B-Jet Analysis

2.3.1 B-Jet Identification Efficiency Objective: Identify signal b-jets from WR^* decay without relying primarily on jet mass reconstruction

Task: Identify AK4 jets originating from signal b-quarks

Geometric Matching Algorithm Truth matching procedure: 1. Use LHE b-quarks as ground truth for signal b identification 2. Find closest AK4 jet to each LHE b-quark using ΔR matching

Distance criteria observed: - **Primary matches:** Closest AK4 typically within $\Delta R < 0.4$ (majority of cases) - **Secondary jets:** Second-closest jets predominantly fall outside $\Delta R > 0.4$ - **Matching reliability:** LHE-to-closest-AK4 matching appears robust for signal identification

Performance Validation Studies B-tagging algorithm comparison (AK4 + AK8 + 2 leptons):

DeepJet algorithm [2022, Tight, 0.7183]: - Efficiency: 8.195% → 8.08% after HLT_IsoMu30 trigger

ParticleNet algorithm [2022, Tight, 0.6734]: - Efficiency: 8.63% → 8.465% after HLT_IsoMu30 trigger

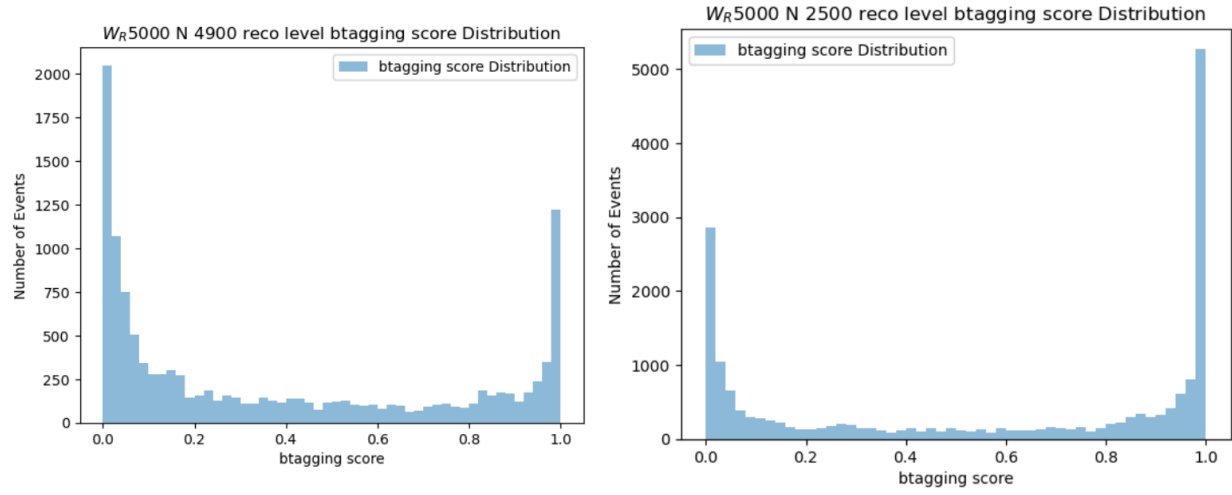


Figure 28: Figure 28: B-tagging performance

Combined selection efficiency (2 AK8 + 2 leptons): - Overall efficiency: 26.9% → 26.375% after HLT_IsoMu30 trigger

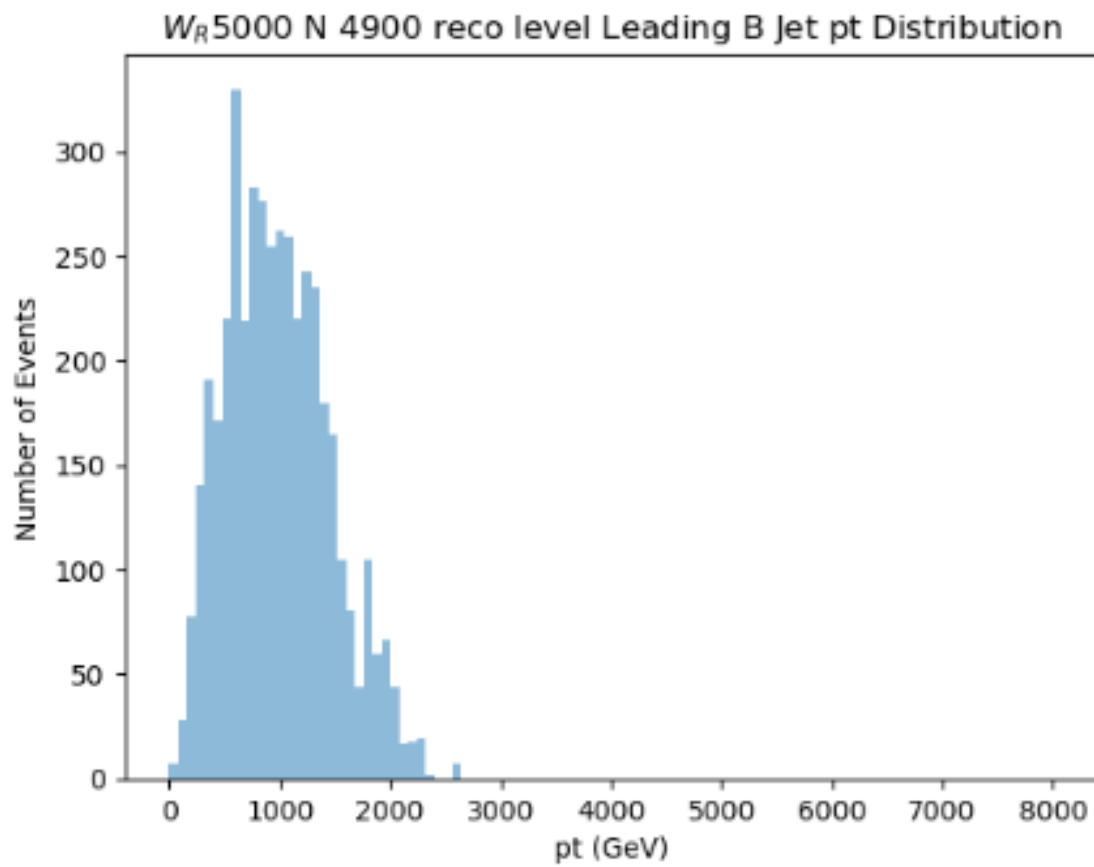


Figure 29: Figure 29: B-jet cleaning results

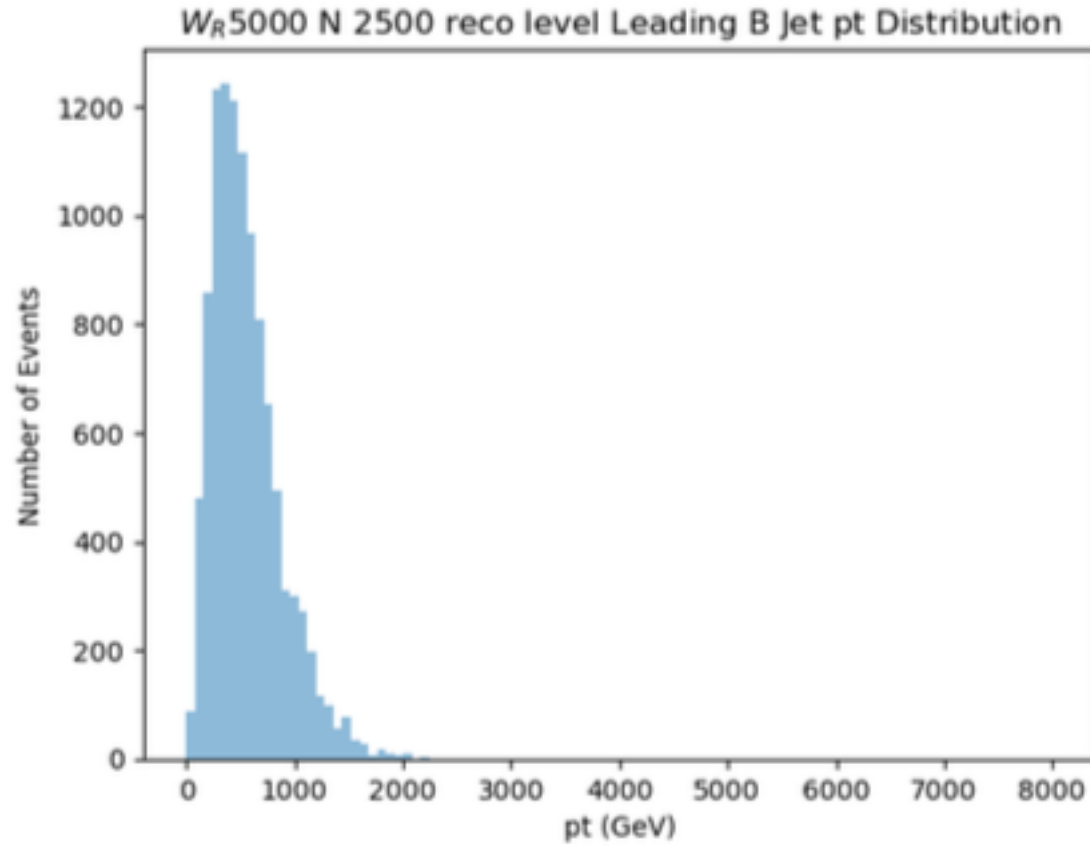


Figure 30: Figure 30: Additional b-jet results

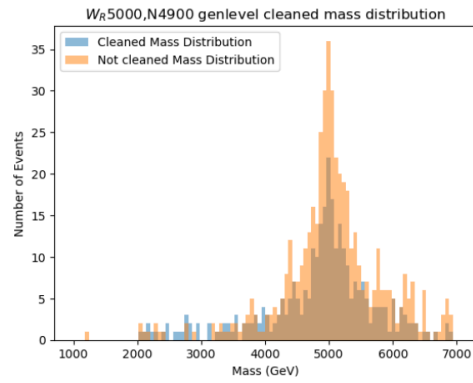
2.3.2 After cleaning

2.3.3 WR Mass Reconstruction Studies Reconstruction scenarios tested for
 $W_R = 5000$, $M_N = 2500$ GeV:

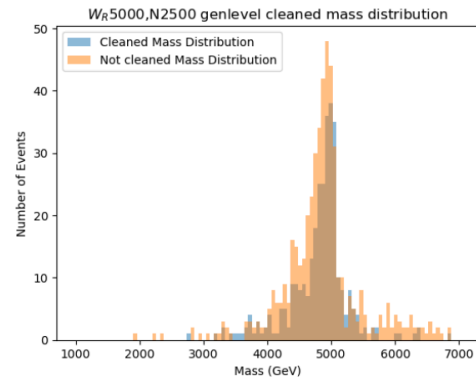
Same criteria selection in gen level

2 leptons , 2AK8

Not used top tagging , b tagging



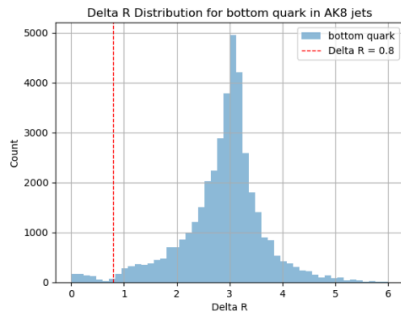
83% efficiency



86% efficiency

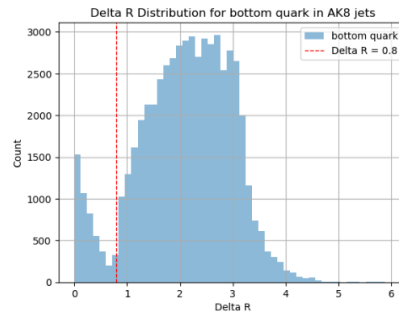
Figure 31: Figure 31: Mass reconstruction comparison

B quark get outside top jet



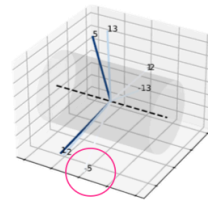
W_R 5000 N 4900

Number of W_R mother bottom quarks with Delta R < 0.8: 678
 Number of t mother bottom quarks outside of AK8: 3303
 Total events processed: 39968



W_R 5000 N 2500

Number of W_R mother bottom quarks with Delta R < 0.8: 4710
 Number of t mother bottom quarks outside of AK8: 12591
 Total events processed: 56759



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Figure 32: Figure 32: W_R mass reconstruction

2.4 Lepton Analysis

Number of reco muon

```
Event 0 has 5 muons with pt: [551, 103, 47.3, 28.5, 27.7]
Event 1 has 7 muons with pt: [567, 370, 334, 51.9, 28.2, 21.7, 12.4]
Event 2 has 3 muons with pt: [924, 67.1, 15.5]
Event 3 has 10 muons with pt: [1.99e+03, 279, 253, 226, 96, 89.9, 57.3, 22.9, 13.2, 3.7]
Event 4 has 5 muons with pt: [519, 397, 80.2, 79.9, 27]
Event 5 has 4 muons with pt: [724, 301, 29.2, 16.8]
Event 6 has 5 muons with pt: [1.51e+03, 906, 35.6, 27.4, 24.5]
Event 7 has 7 muons with pt: [452, 219, 64.1, 31, 29, 17.3, 16.2]
Event 8 has 4 muons with pt: [557, 147, 18.7, 15.3]
Event 9 has 4 muons with pt: [1.11e+03, 83.4, 47.5, 3.5]
Event 10 has 4 muons with pt: [969, 121, 7.78, 5.04]
Event 11 has 4 muons with pt: [1.06e+03, 121, 29.6, 26.1]
Event 12 has 5 muons with pt: [534, 423, 21.2, 20.8, 4.64]
Event 13 has 3 muons with pt: [834, 258, 7.33]
Event 14 has 5 muons with pt: [545, 423, 13.6, 9.88, 5.85]
Event 15 has 5 muons with pt: [1.88e+03, 995, 162, 92.6, 15.6]
Event 16 has 8 muons with pt: [244, 88.8, 68.4, 66.7, 45.7, 38.9, 19.2, 6.26]
Event 17 has 7 muons with pt: [552, 254, 116, 28.5, 27.2, 23.4, 21.6]
Event 18 has 2 muons with pt: [158, 123]
Event 19 has 3 muons with pt: [1.48e+03, 874, 230]
Event 20 has 10 muons with pt: [301, 282, 193, 99.2, 75.8, 48.4, 23.5, 22, 20.4, 15.9]
Event 21 has 2 muons with pt: [462, 192]
Event 22 has 2 muons with pt: [2.4e+03, 190]
Event 23 has 2 muons with pt: [89.8, 15.8]
Event 24 has 9 muons with pt: [805, 621, 63.1, 52.9, 51.5, 48.6, 29.9, 11.3, 10.4]
...
Event 497 has 3 muons with pt: [1.93e+03, 197, 5.15]
Event 498 has 2 muons with pt: [959, 176]
Event 499 has 5 muons with pt: [632, 308, 114, 23.7, 22.2]
5.647294589178356 average muon number per event
```

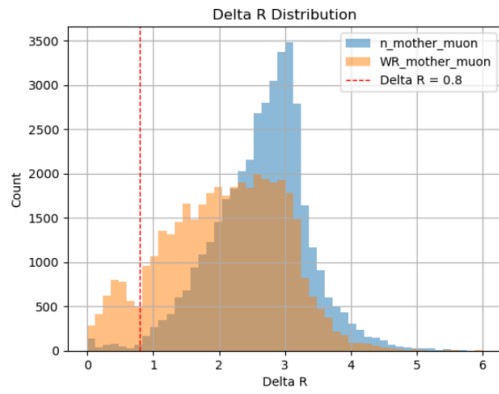
W_R 5000 , N 4900

```
Event 0 has 2 muons with pt: [1.19e+03, 779]
Event 1 has 3 muons with pt: [745, 640, 4.5]
Event 2 has 7 muons with pt: [1.58e+03, 1.02e+03, 328, 120, 114, 40.9, 8.53]
Event 3 has 6 muons with pt: [1.67e+03, 673, 40.6, 36.9, 31.8, 25.1]
Event 4 has 6 muons with pt: [1.07e+03, 513, 26.3, 25, 18.5, 13.9]
Event 5 has 4 muons with pt: [1.15e+03, 865, 137, 64.5]
Event 6 has 2 muons with pt: [863, 628]
Event 7 has 3 muons with pt: [1.67e+03, 628, 4.7]
Event 8 has 4 muons with pt: [1.13e+03, 881, 5.45, 5.3]
Event 9 has 3 muons with pt: [1.36e+03, 255, 9.25]
Event 10 has 4 muons with pt: [1.93e+03, 747, 5.08, 3.96]
Event 11 has 4 muons with pt: [1.25e+03, 1.09e+03, 89.7, 44.1]
Event 12 has 7 muons with pt: [1.78e+03, 314, 89, 49.2, 46.6, 34.4, 23.8]
Event 13 has 5 muons with pt: [1.71e+03, 1.53e+03, 609, 41.1, 32.6]
Event 14 has 4 muons with pt: [1.9e+03, 427, 28.5, 10.2]
Event 15 has 2 muons with pt: [1.63e+03, 1.16e+03]
Event 16 has 3 muons with pt: [1.19e+03, 268, 4.62]
Event 17 has 3 muons with pt: [1.42e+04, 1.43e+03, 602]
Event 18 has 4 muons with pt: [1.69e+03, 1.39e+03, 10.7, 7.6]
Event 19 has 4 muons with pt: [1.4e+03, 243, 53.2, 6.67]
Event 20 has 3 muons with pt: [1.75e+03, 373, 21.8]
Event 21 has 4 muons with pt: [1.67e+03, 1.45e+03, 8.4, 4.85]
Event 22 has 2 muons with pt: [1.48e+03, 1.02e+03]
Event 23 has 3 muons with pt: [971, 650, 152]
Event 24 has 4 muons with pt: [1.15e+03, 476, 106, 69.1]
...
Event 497 has 4 muons with pt: [743, 666, 366, 3.27]
Event 498 has 7 muons with pt: [694, 243, 38.1, 27.2, 17.1, 13.8, 3.72]
Event 499 has 5 muons with pt: [1.04e+03, 944, 18.8, 15.8, 4.04]
4.040886160320641 average muon number per event
```

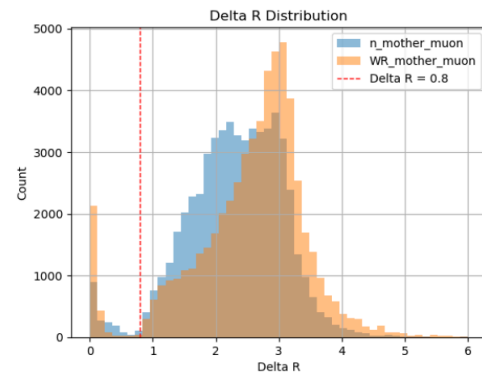
W_R 5000 , N 2500

Figure 33: Figure 33: Muon features

Number of muon inside top jets



W_R 5000 , N 4900



W_R 5000 , N 2500

Gets inside $\sim 10\%$

Figure 34: Figure 34: Additional muon properties

Muon p_t distribution with signal & bkg

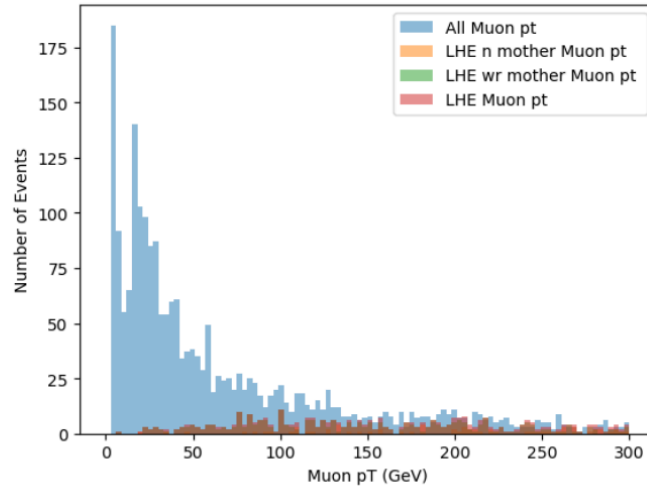


Figure 35: Figure 35: Muon distributions

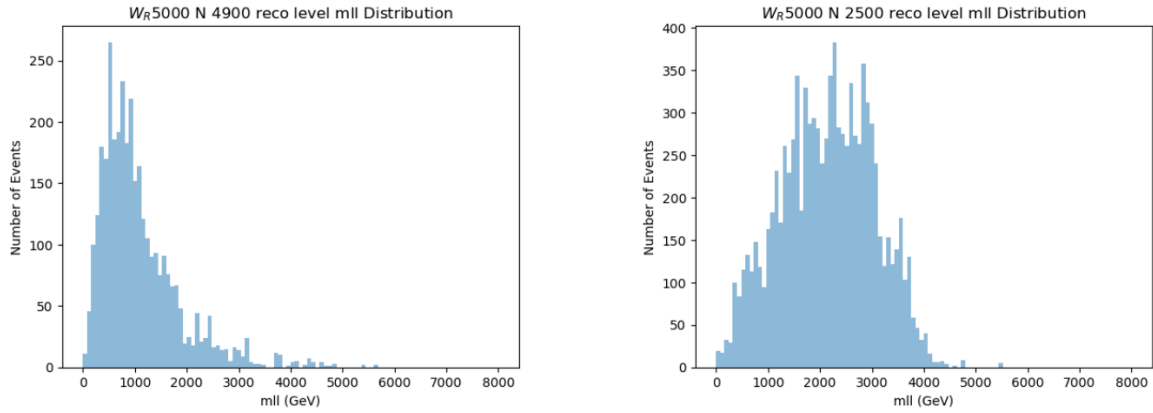


Figure 36: Figure 36: Muon analysis

2.4.1 Muon features

2.4.2 Lepton cleaning Cases where they entered this ak8 (b jet overlap cases were not many in 5000 4900): → It seems we can use leptons from lhe and use the found fatjet.

WR 5000 N2500: survived jet: 52313 // cleaned jet: 4410 // total jet: 56723 @ On average, 4 muons are contained.

WR 5000 N 4900: survived jet: 36003 // cleaned jet: 3953 // total jet: 39956 @ On average, 6.5 muons are contained.

2.4.3 Lepton ID, Trigger selection To get Leptons, first it should go through HLT, and we should choose leptons with ID.

For Muon, checking features first:

1. pt

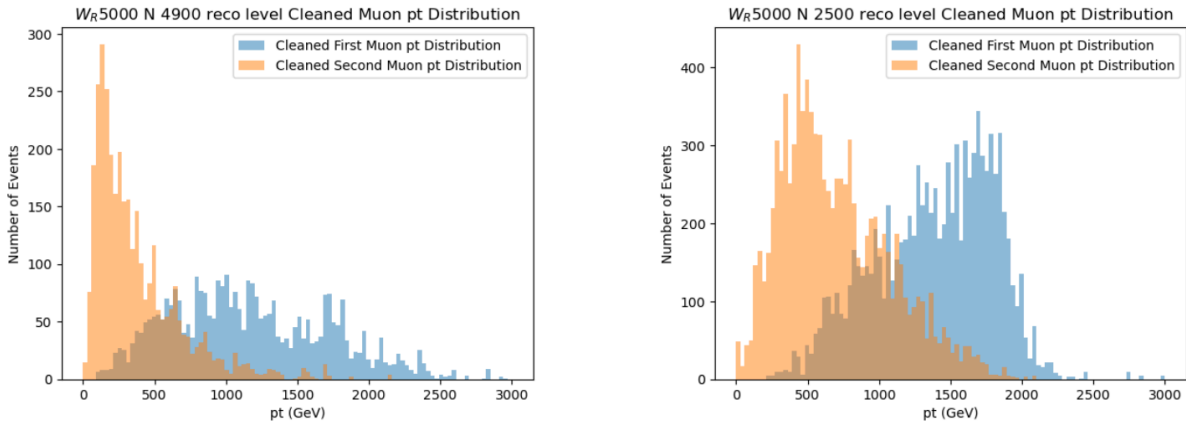


Figure 37: Figure 37: Muon pT

2. High pt id and isolated

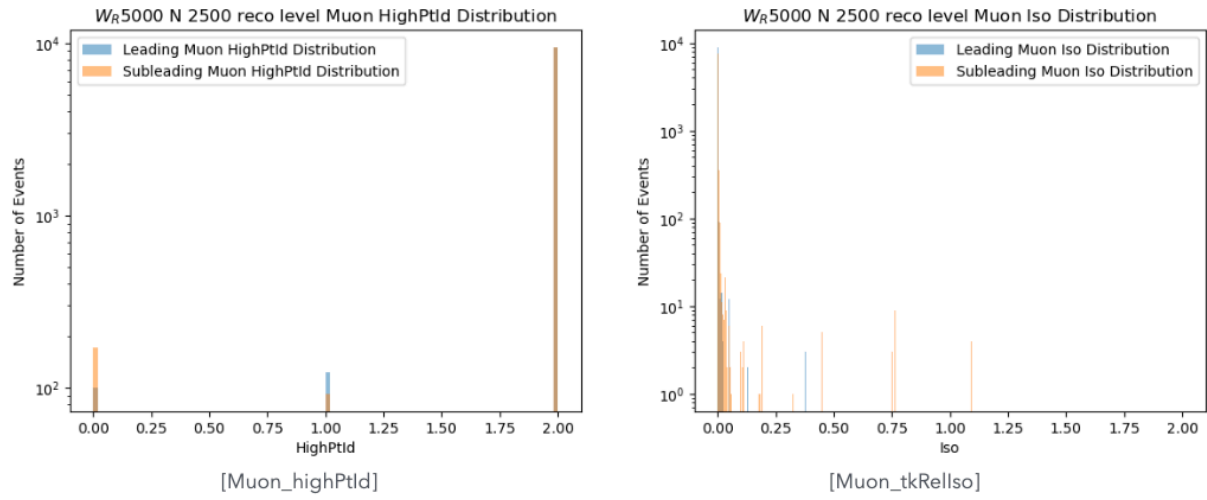


Figure 38: Figure 38: High pT ID

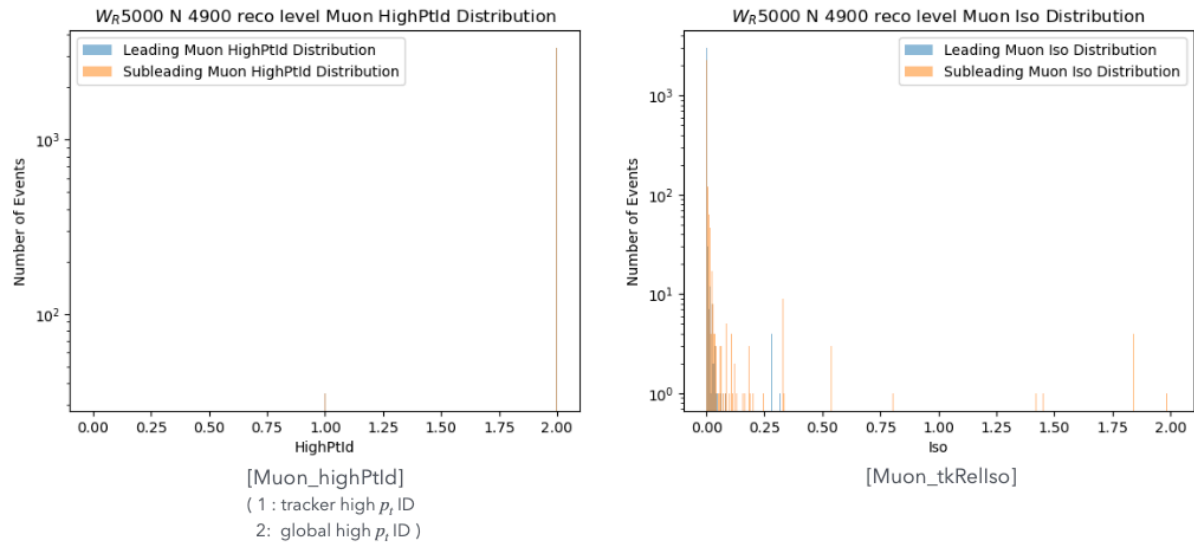


Figure 39: Figure 39: Isolation

which mean we can use high pt id, and iso id, must also check recommendation.

HLT recommendation gives:

Event type	Recommended path	Comments
Single muon, intermediate pT	HLT_IsoMu24_v*	- Unprescaled for the whole year - final filter: hlt3cris0L1sSingleMu22L1f0L2f10QL3f24QL3trkIsoFiltered
Single muon, high pT	HLT_Mu50_v* OR HLT_CascadeMu100_v* OR HLT_HighPTkMu100_v*	- Unprescaled for the whole year - final filters: hlt3fL1sMu22Or25L1f0L2f10QL3Filtered50Q, hlt3fL1sMu22Or25L1f0L2f10QL3Filtered100Q, hlt3fL1sMu25f0TkFiltered100Q - OR with CascadeMu100 & HighPTkMu100 : recover the inefficiency on high pT (~O(100)GeV) muons due to global chi2ndof cut in Mu50 trigger (details: slides.c)
Double muon, intermediate pT	HLT_Mu17_TrkIsoVVL_Mu8_TrkIsoVVL_DZ_Mass3p8_v*	- Unprescaled for the whole year

Figure 40: Figure 40: HLT recommendations

And for ID's:

Particle-Flow (PF) Algorithm

- 전체 이벤트의 글로벌 정보를 활용하여 모든 입자(charged hadrons, neutral hadrons, photons, electrons, muons)를 종합적으로 재구성
- 각 입자의 에너지와 운동량을 모든 서브검출기 정보를 조합하여 최적화
- 이벤트 레벨에서의 일관성을 보장

Non-PF (Traditional) Approach

- 뮤온 검출기와 트랙커 정보만을 사용하여 뮤온을 재구성
- 다른 입자들과의 상호작용을 고려하지 않음
- 독립적인 뮤온 재구성에 집중

2. 주요 차이점

특징	PF-based (Tight ID)	Non-PF-based (High-pT ID)
적용 범위	pT < 200 GeV 권장	pT > 200 GeV 전용
알고리즘	Particle-Flow 종합 재구성	TuneP 알고리즘 사용
정보 활용	전체 이벤트 정보	뮤온 검출기 + 트랙커만
효율성	pT > 200 GeV에서 효율성 저하	고에너지에서 효율성 유지
가짜 비율	낮은 fake rate	상대적으로 높은 fake rate
운동량 할당	PF 기반 최적화	TuneP 기반 최적화

Figure 41: Figure 41: Muon ID recommendations

Conclusion: For muon selection, use: - **HLT:** HLT_Mu50 || HLT_HighPtTkMu100 || HLT_CascadeMu100 due to pt over 1 TeV - **ID:** HighptID == 2 (tight), Muon_tkIsoId == 2 (tight) - **pT cut:** starts from 50 GeV for SF availability

High pT: ID and ISO efficiencies

Identification

- The ID working points provided are: Medium, Tight, High-pT and Tracker High-pT IDs.
- **Note:** Tight ID efficiency drops toward high-pT values, so **it is not recommended**.

Isolation

- The isolation working points are: Loose Relative Tracker Isolation over Medium, High-pT and Tracker High-pT IDs, and Tight Relative Tracker Isolation over the same ID working points.

Figure 42: Figure 42: Final muon selection

3. Background Analysis

Selection cuts comparison:

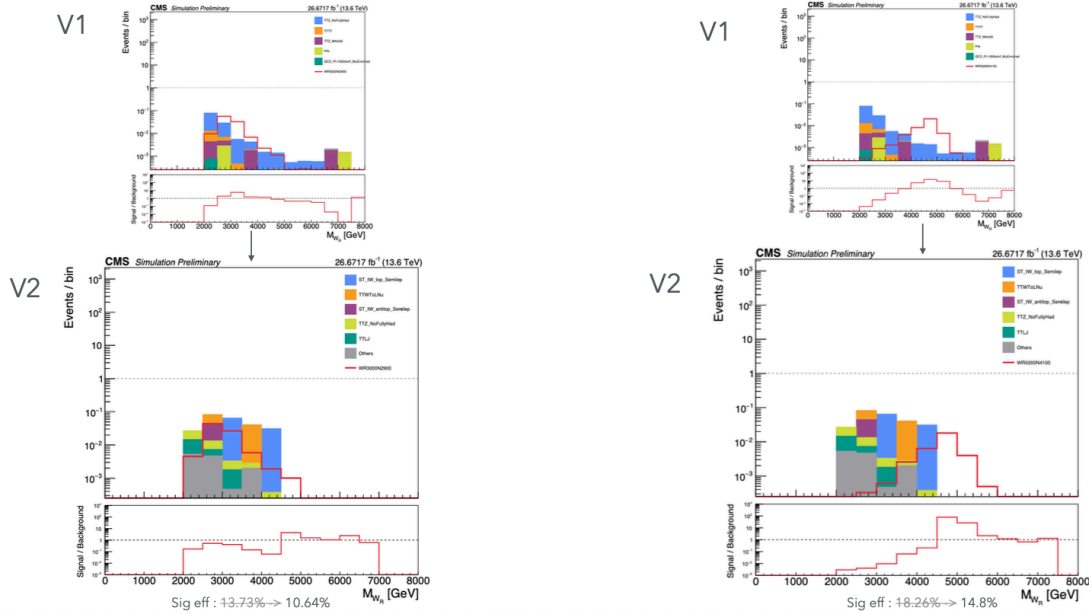
Cut	lead HLT Muon pt	sub lead muon pt	B Bjet pt	Top score	Top SDM	Top pt	WR mass	m(ll)
cut1	ISOMu30	0	Tight 30	0.9	120 250	0	2000	300
cut2	ISOMu30	0	Tight 300	0.9	120 250	500	2000	300

cut1: BKG → TTHtononbb, TTLJ, TTTT, TTWtonu, TTZ, TZq

cut2: BKG → ST_TW_antitop_Semilep, ST_tW_top_Semilep, TTG_PTG200toinf, TTHToNonbb, TTLJ, TTTT, TTWtoNu, TTZ_M4to50, TTZ_M50, TTZ_NoFullyHad

Selection cuts v.2 results

[2022 E.F.G era]



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Figure 43: Figure 43: Background analysis

Issues with these cuts:

1. Muon trigger is not a recommended one, also it is isolated and high pt, which can use other HLT
2. No Muon ID selected, also pt cut must be chosen with Muon ID min pt
3. For cut2 B pt is too high to adjust scale factors. It should have some pt to measure scale factor which we can observe in data
4. Also mll cut or wr mass cut gives large constraints of background, which makes it not available to measure SF which is the discrepancy of data and signal

Summary and Conclusions

This analysis presents a comprehensive study of the LRSM $WR \rightarrow tb$ channel, focusing on the $WR \sim N$ mass region to optimize signal reconstruction and background rejection. Key findings include:

1. **Kinematic Understanding:** The $WR \sim N$ mass region provides better separation of decay products compared to the highly boosted regime

2. **Top Jet Reconstruction:** AK8 jets with ParticleNet top tagging achieve ~55-40% efficiency after cleaning
3. **B-jet Identification:** AK4 b-jets can be identified with moderate efficiency using geometric matching and flavor tagging
4. **Lepton Selection:** High-pT muon triggers and tight ID requirements are essential for signal selection
5. **Background Challenges:** Careful optimization of selection cuts is needed to maintain sensitivity while allowing for scale factor measurements

Future work should focus on optimizing the selection criteria and developing data-driven background estimation methods.

Appendix A: Technical Details

A.1 Software and Analysis Framework

Analysis Tools: - MadGraph for signal generation - ROOT framework for data analysis
- NanoAOD format for event storage - ParticleNet for top/b-tagging

A.2 Selection Criteria Summary

Final Recommended Cuts:

Object	Selection Criteria	Efficiency
Muons	$p_T > 50 \text{ GeV}$, HighPtID=2, tkIsoId=2	~90%
AK8 Jets	$p_T > 200 \text{ GeV}$, ParticleNet > 0.9 , 120 $< \text{SDM} < 250 \text{ GeV}$	~55-40%
AK4 b-jets	$p_T > 30 \text{ GeV}$, DeepJet Tight WP	~8%
HLT	HLT_Mu50 OR HLT_HighPtTkMu100 OR HLT_CascadeMu100	~95%

A.3 Systematic Uncertainties

Expected Sources: - Jet Energy Scale/Resolution - b-tagging scale factors - Top-tagging scale factors - Muon ID/trigger scale factors - Luminosity uncertainty - PDF uncertainties

A.4 Background Composition

Dominant Backgrounds: - $t\bar{t}$ + jets (dominant) - Single top - $t\bar{t}$ + V (V = W, Z) - $t\bar{t}$ + H - Diboson production

A.5 Signal Optimization Strategy

Mass-dependent Approach: - WR $\sim$ N region: Resolved analysis with separated objects - High WR region: Boosted analysis with merged jets - Transition region: Hybrid approach

References

1. CMS Collaboration, “Search for heavy right-handed W bosons in events with an electron and missing transverse momentum”(EXO-20-002)
 2. CMS Top Tagging Working Group recommendations
 3. CMS B-tagging Performance Studies, Summer22
 4. CMS Muon Performance Studies, Run 3
 5. Left-Right Symmetric Model theoretical framework
-

Document prepared for A4 PDF conversion with optimized page breaks and image placement